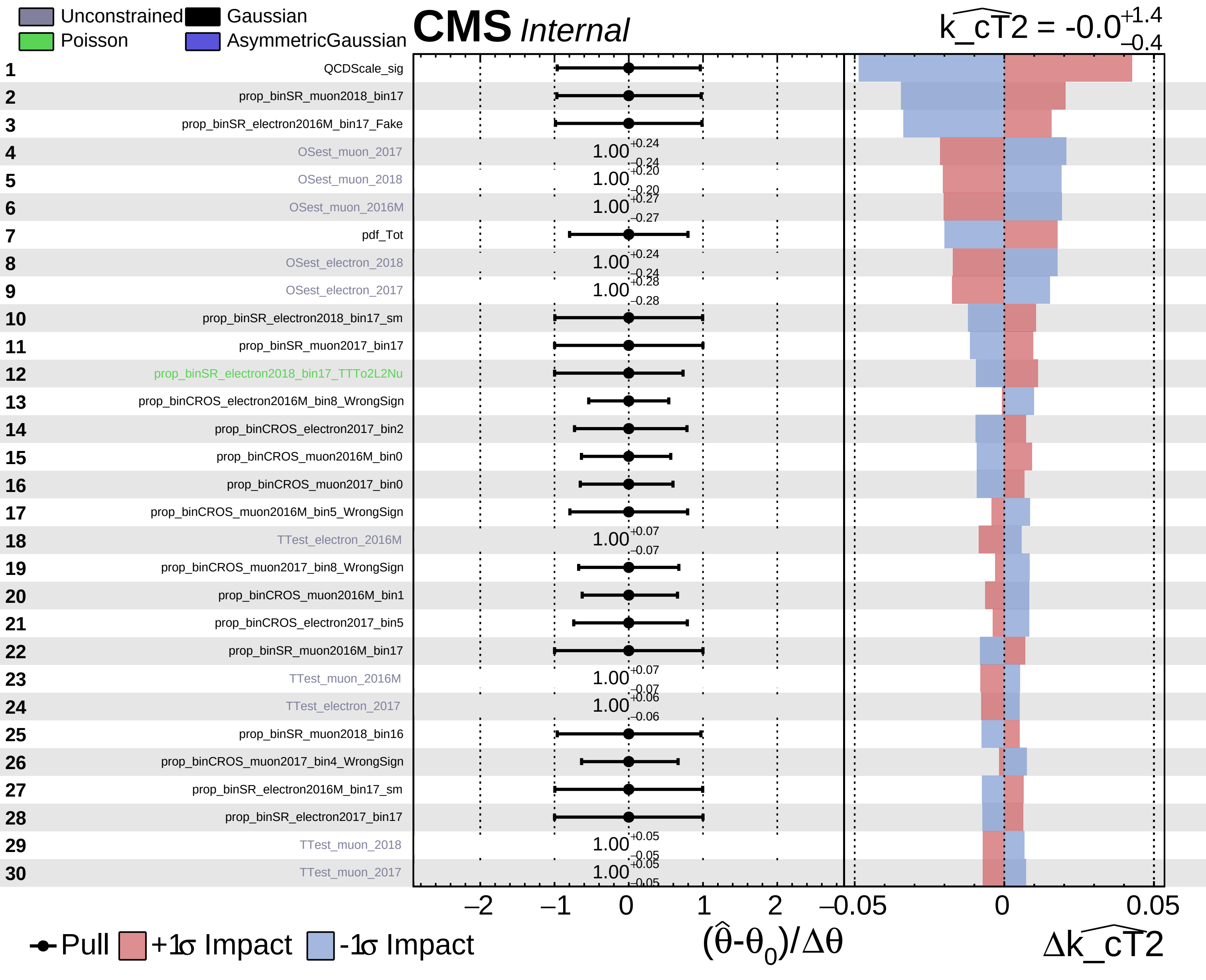
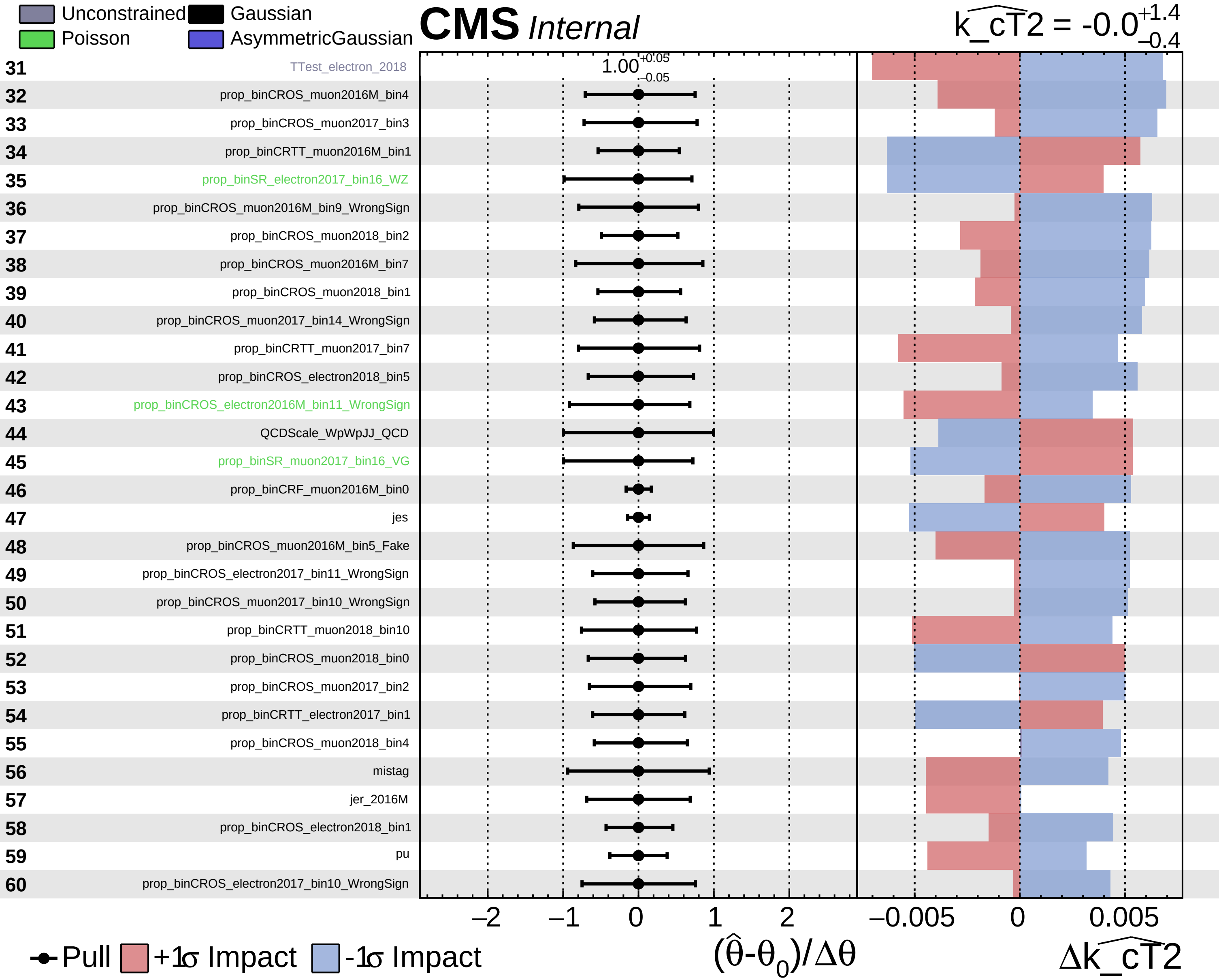
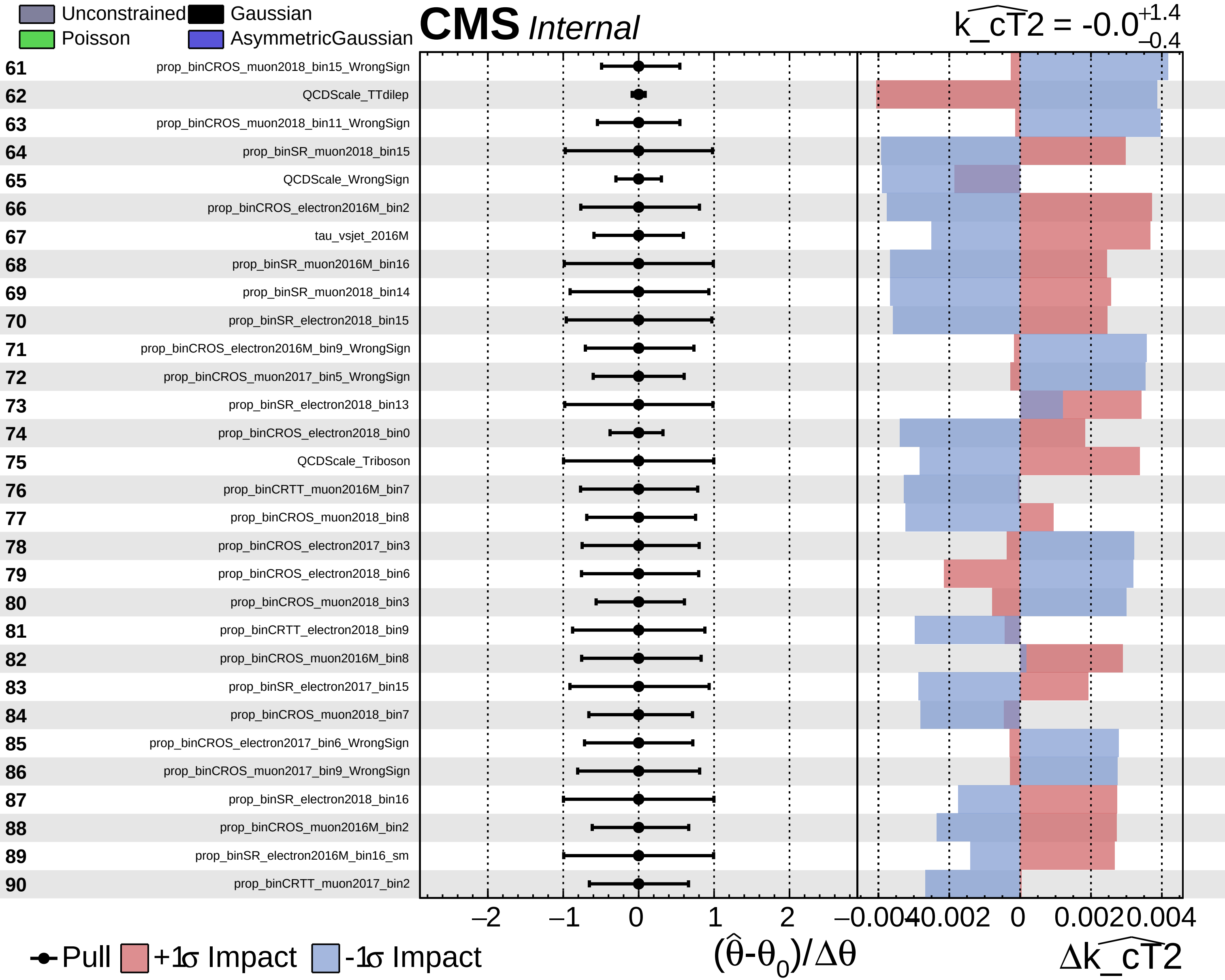


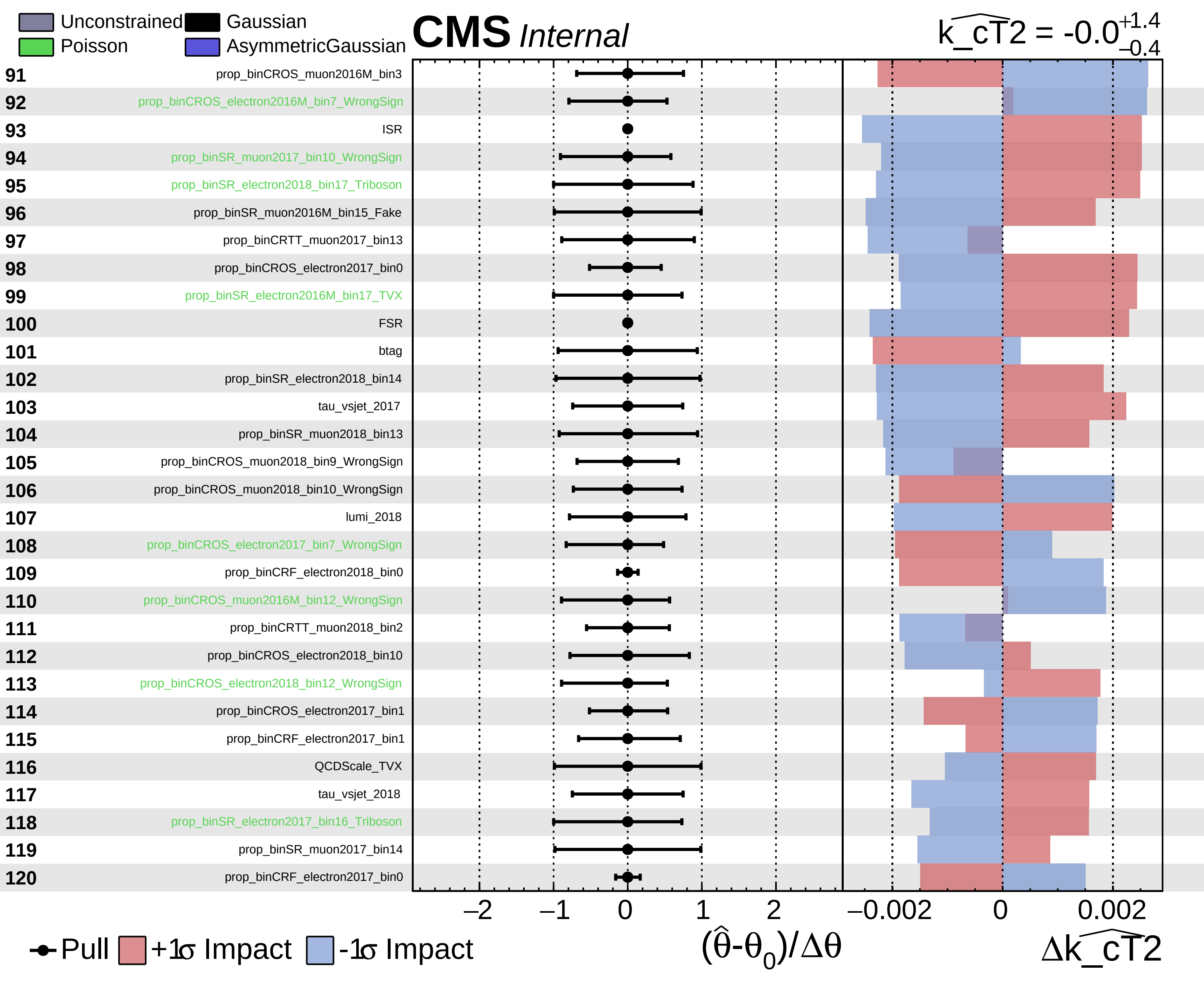
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +1.4 −0.4
1 QCDScale_sig
2 prop_binSR_muon2018_bin17
3 prop_binSR_electron2016M_bin17_Fake
4 OSest_muon_2017 1.00 +0.24 −0.24
5 OSest_muon_2018 1.00 +0.20 −0.20
6 OSest_muon_2016M 1.00 +0.27 −0.27
7 pdf_Tot
8 OSest_electron_2018 1.00 +0.24 −0.24
9 OSest_electron_2017 1.00 +0.28 −0.28
10 prop_binSR_electron2018_bin17_sm
11 prop_binSR_muon2017_bin17
12 prop_binSR_electron2018_bin17_TTTo2L2Nu
13 prop_binCROS_electron2016M_bin8_WrongSign
14 prop_binCROS_electron2017_bin2
15 prop_binCROS_muon2016M_bin0
16 prop_binCROS_muon2017_bin0
17 prop_binCROS_muon2016M_bin5_WrongSign
18 TTest_electron_2016M 1.00 +0.07 −0.07
19 prop_binCROS_muon2017_bin8_WrongSign
20 prop_binCROS_muon2016M_bin1
21 prop_binCROS_electron2017_bin5
22 prop_binSR_muon2016M_bin17
23 TTest_muon_2016M 1.00 +0.07 −0.07
24 TTest_electron_2017 1.00 +0.06 −0.06
25 prop_binSR_muon2018_bin16
26 prop_binCROS_muon2017_bin4_WrongSign
27 prop_binSR_electron2016M_bin17_sm
28 prop_binSR_electron2017_bin17
29 TTest_muon_2018 1.00 +0.05 −0.05
30 TTest_muon_2017 1.00 +0.05 −0.05
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



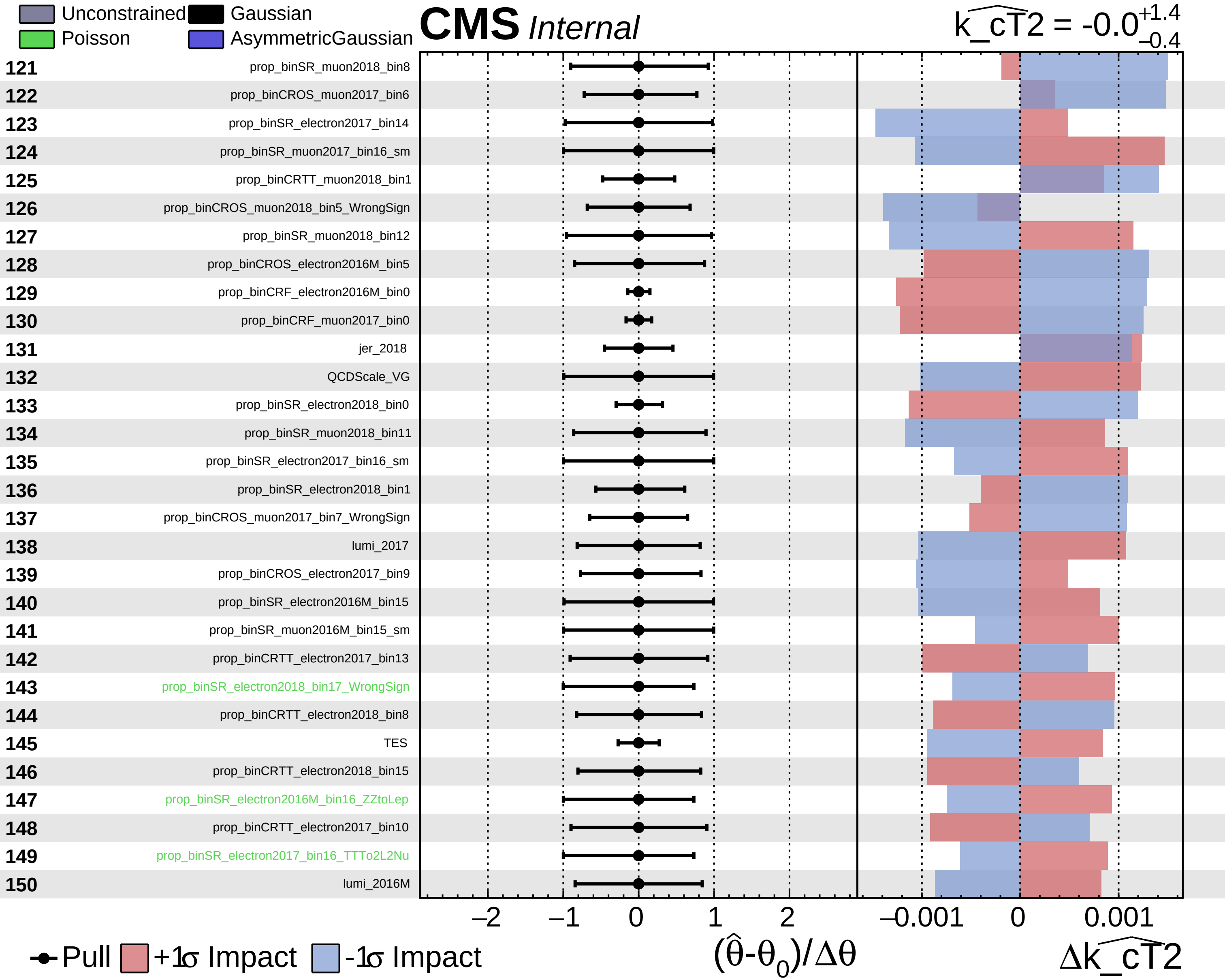
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +1.4 −0.4
1.00 +0.05 −0.05
31 TTest_electron_2018
32 prop_binCROS_muon2016M_bin4
33 prop_binCROS_muon2017_bin3
34 prop_binCRTT_muon2016M_bin1
35 prop_binSR_electron2017_bin16_WZ
36 prop_binCROS_muon2016M_bin9_WrongSign
37 prop_binCROS_muon2018_bin2
38 prop_binCROS_muon2016M_bin7
39 prop_binCROS_muon2018_bin1
40 prop_binCROS_muon2017_bin14_WrongSign
41 prop_binCRTT_muon2017_bin7
42 prop_binCROS_electron2018_bin5
43 prop_binCROS_electron2016M_bin11_WrongSign
44 QCDScale_WpWpJJ_QCD
45 prop_binSR_muon2017_bin16_VG
46 prop_binCRF_muon2016M_bin0
47 jes
48 prop_binCROS_muon2016M_bin5_Fake
49 prop_binCROS_electron2017_bin11_WrongSign
50 prop_binCROS_muon2017_bin10_WrongSign
51 prop_binCRTT_muon2018_bin10
52 prop_binCROS_muon2018_bin0
53 prop_binCROS_muon2017_bin2
54 prop_binCRTT_electron2017_bin1
55 prop_binCROS_muon2018_bin4
56 mistag
57 jer_2016M
58 prop_binCROS_electron2018_bin1
59 pu
60 prop_binCROS_electron2017_bin10_WrongSign
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



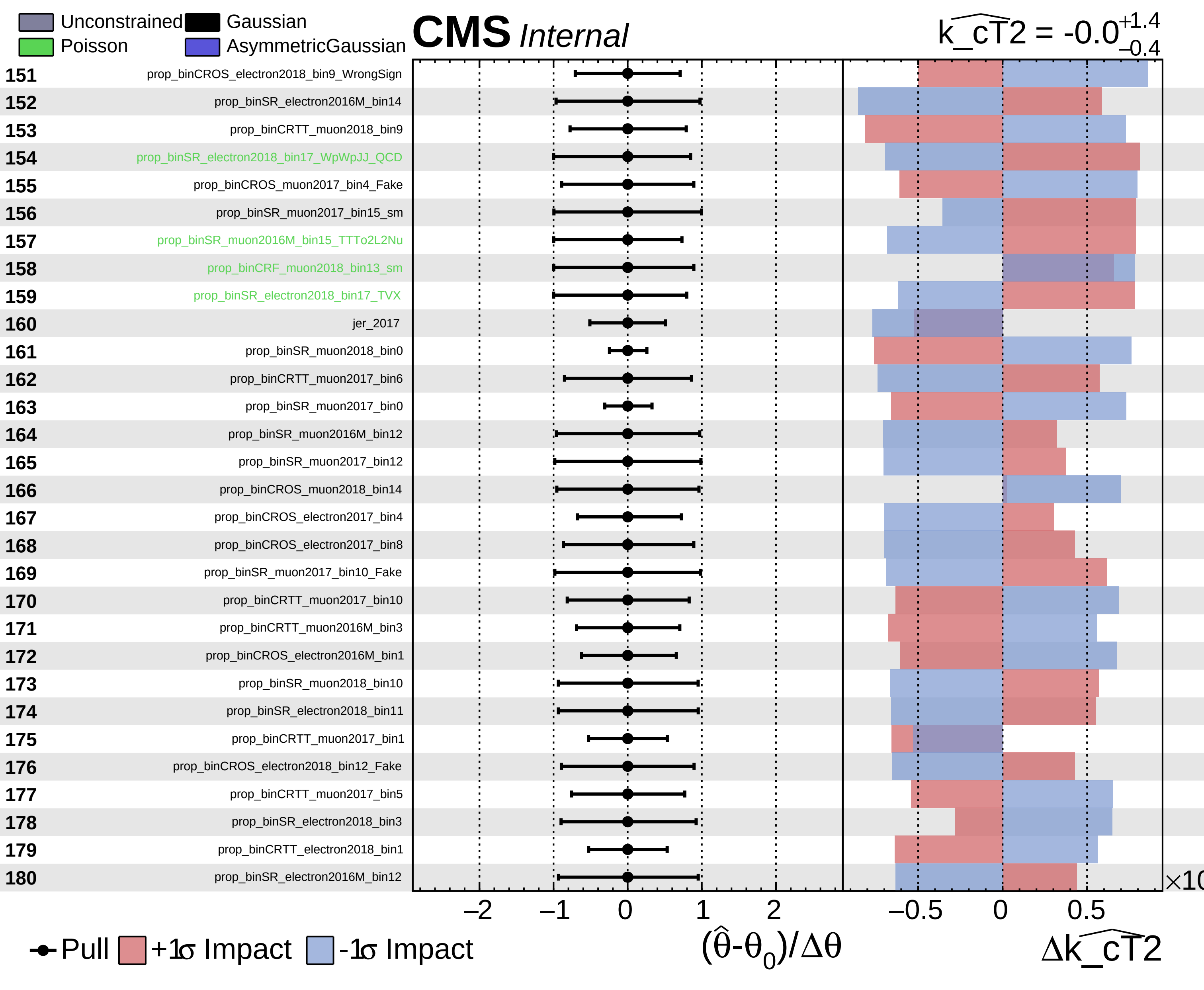
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+1.4}_{-0.4}$
61 prop_binCROS_muon2018_bin15_WrongSign
62 QCDScale_TTdilep
63 prop_binCROS_muon2018_bin11_WrongSign
64 prop_binSR_muon2018_bin15
65 QCDScale_WrongSign
66 prop_binCROS_electron2016M_bin2
67 tau_vsjet_2016M
68 prop_binSR_muon2016M_bin16
69 prop_binSR_muon2018_bin14
70 prop_binSR_electron2018_bin15
71 prop_binCROS_electron2016M_bin9_WrongSign
72 prop_binCROS_muon2017_bin5_WrongSign
73 prop_binSR_electron2018_bin13
74 prop_binCROS_electron2018_bin0
75 QCDScale_Triboson
76 prop_binCRTT_muon2016M_bin7
77 prop_binCROS_muon2018_bin8
78 prop_binCROS_electron2017_bin3
79 prop_binCROS_electron2018_bin6
80 prop_binCROS_muon2018_bin3
81 prop_binCRTT_electron2018_bin9
82 prop_binCROS_muon2016M_bin8
83 prop_binSR_electron2017_bin15
84 prop_binCROS_muon2018_bin7
85 prop_binCROS_electron2017_bin6_WrongSign
86 prop_binCROS_muon2017_bin9_WrongSign
87 prop_binSR_electron2018_bin16
88 prop_binCROS_muon2016M_bin2
89 prop_binSR_electron2016M_bin16_sm
90 prop_binCRTT_muon2017_bin2
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



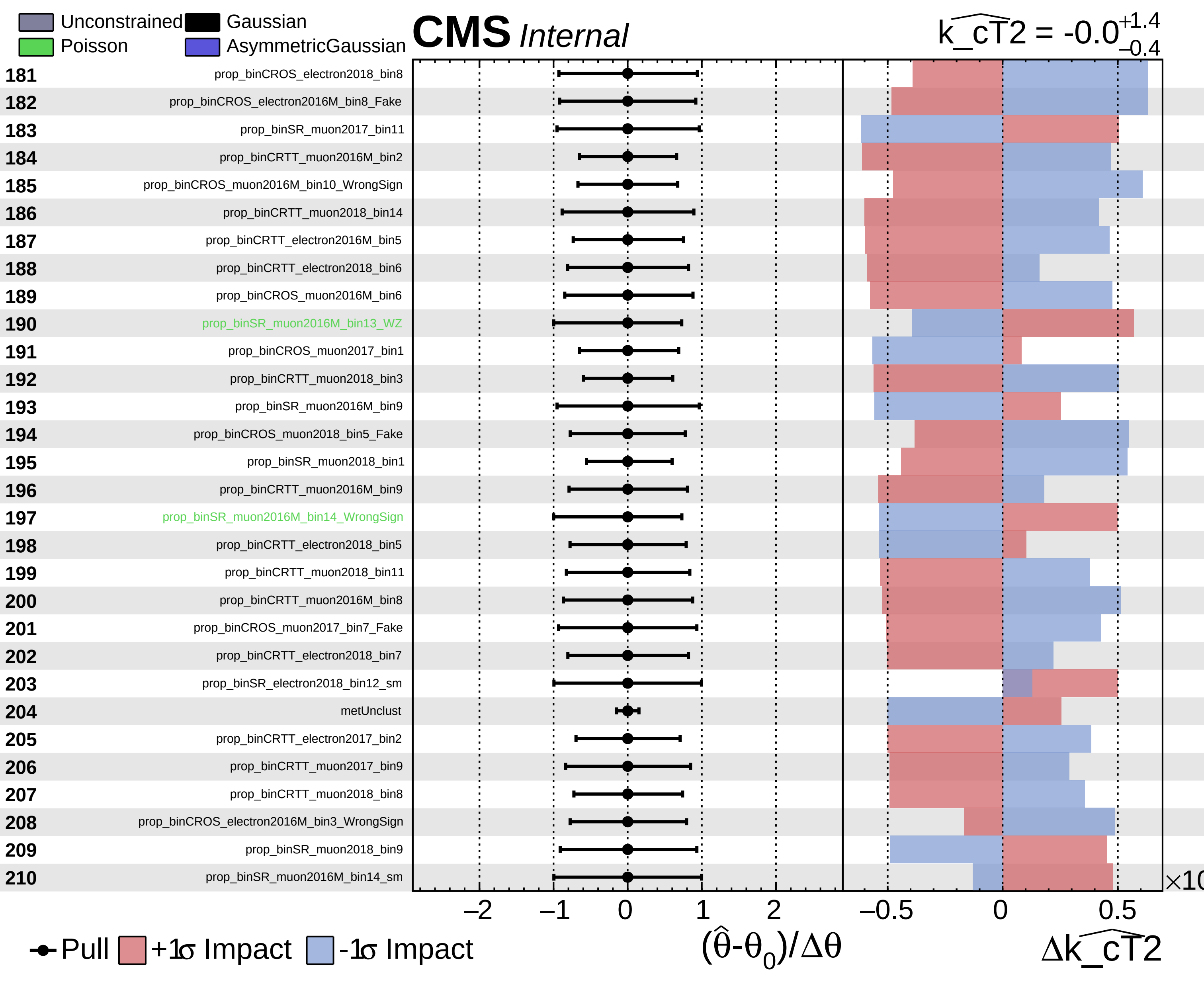
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+1.4}_{-0.4}$
91 prop_binCROS_muon2016M_bin3
92 prop_binCROS_electron2016M_bin7_WrongSign
93 ISR
94 prop_binSR_muon2017_bin10_WrongSign
95 prop_binSR_electron2018_bin17_Triboson
96 prop_binSR_muon2016M_bin15_Fake
97 prop_binCRTT_muon2017_bin13
98 prop_binCROS_electron2017_bin0
99 prop_binSR_electron2016M_bin17_TVX
100 FSR
101 btag
102 prop_binSR_electron2018_bin14
103 tau_vsjet_2017
104 prop_binSR_muon2018_bin13
105 prop_binCROS_muon2018_bin9_WrongSign
106 prop_binCROS_muon2018_bin10_WrongSign
107 lumi_2018
108 prop_binCROS_electron2017_bin7_WrongSign
109 prop_binCRF_electron2018_bin0
110 prop_binCROS_muon2016M_bin12_WrongSign
111 prop_binCRTT_muon2018_bin2
112 prop_binCROS_electron2018_bin10
113 prop_binCROS_electron2018_bin12_WrongSign
114 prop_binCROS_electron2017_bin1
115 prop_binCRF_electron2017_bin1
116 QCDScale_TVX
117 tau_vsjet_2018
118 prop_binSR_electron2017_bin16_Triboson
119 prop_binSR_muon2017_bin14
120 prop_binCRF_electron2017_bin0
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



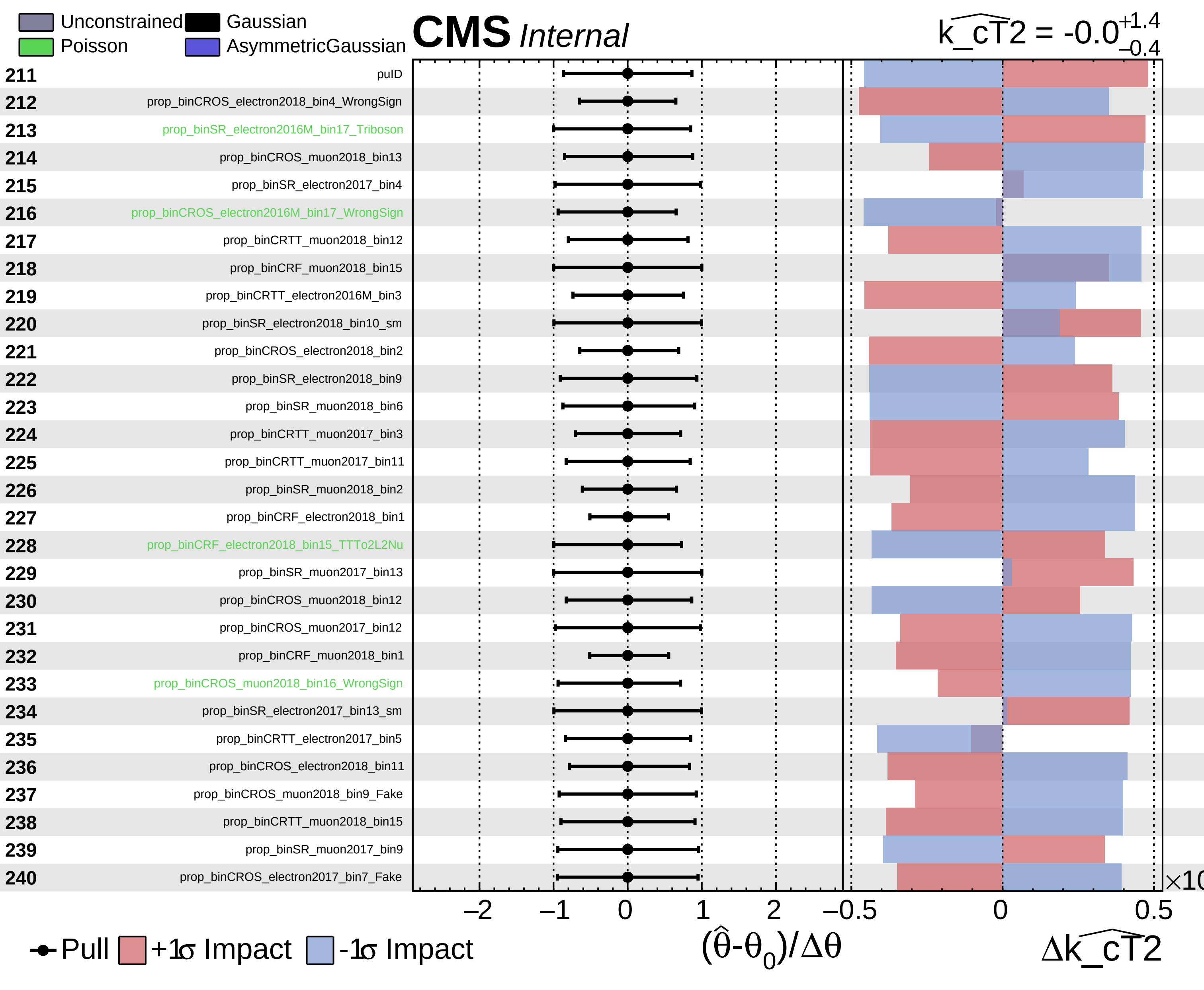
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +1.4 −0.4
121 prop_binSR_muon2018_bin8
122 prop_binCROS_muon2017_bin6
123 prop_binSR_electron2017_bin14
124 prop_binSR_muon2017_bin16_sm
125 prop_binCRTT_muon2018_bin1
126 prop_binCROS_muon2018_bin5_WrongSign
127 prop_binSR_muon2018_bin12
128 prop_binCROS_electron2016M_bin5
129 prop_binCRF_electron2016M_bin0
130 prop_binCRF_muon2017_bin0
131 jer_2018
132 QCDScale_VG
133 prop_binSR_electron2018_bin0
134 prop_binSR_muon2018_bin11
135 prop_binSR_electron2017_bin16_sm
136 prop_binSR_electron2018_bin1
137 prop_binCROS_muon2017_bin7_WrongSign
138 lumi_2017
139 prop_binCROS_electron2017_bin9
140 prop_binSR_electron2016M_bin15
141 prop_binSR_muon2016M_bin15_sm
142 prop_binCRTT_electron2017_bin13
143 prop_binSR_electron2018_bin17_WrongSign
144 prop_binCRTT_electron2018_bin8
145 TES
146 prop_binCRTT_electron2018_bin15
147 prop_binSR_electron2016M_bin16_ZZtoLep
148 prop_binCRTT_electron2017_bin10
149 prop_binSR_electron2017_bin16_TTTo2L2Nu
150 lumi_2016M
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



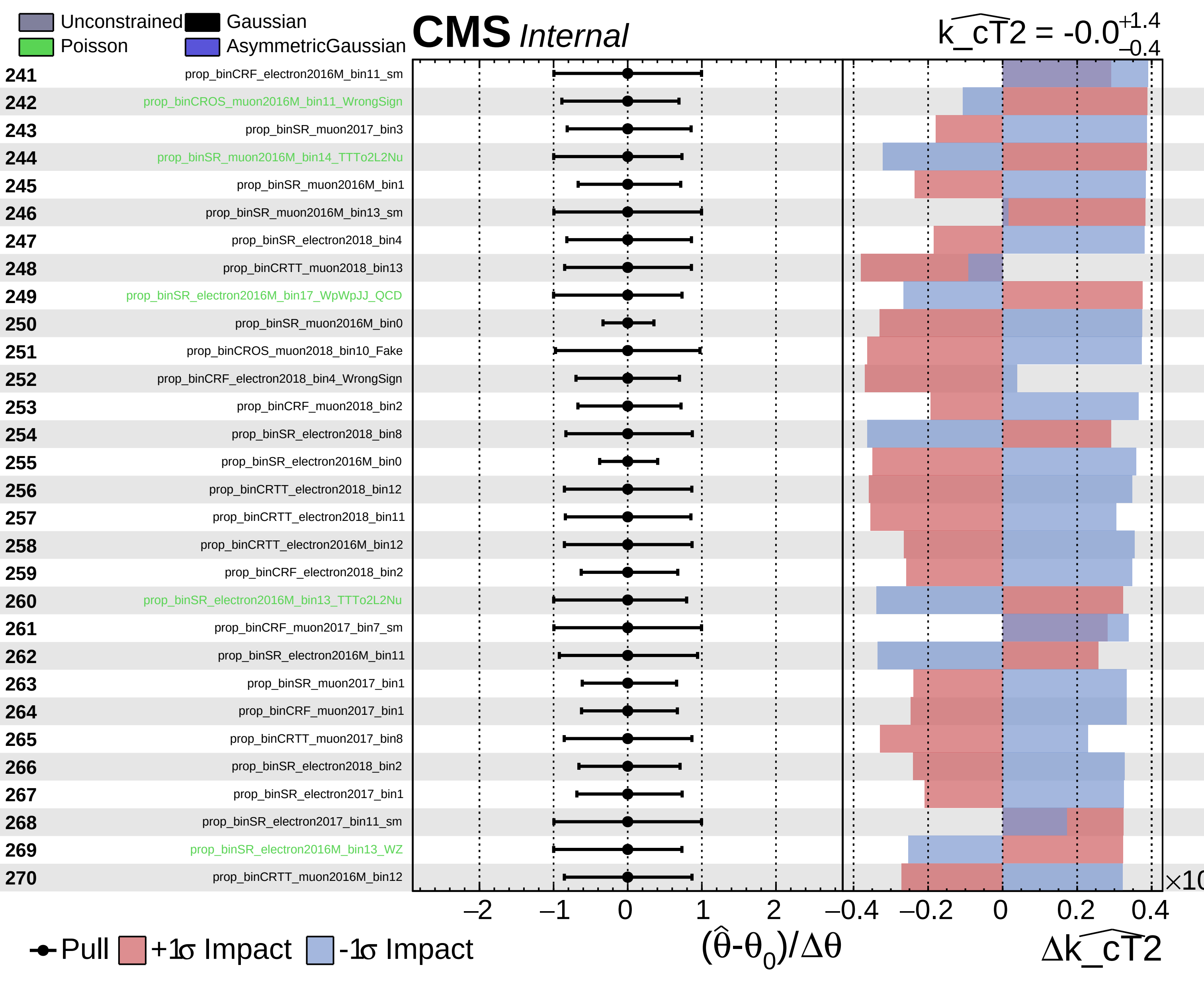
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+1.4}_{-0.4}$
151 prop_binCROS_electron2018_bin9_WrongSign
152 prop_binSR_electron2016M_bin14
153 prop_binCRTT_muon2018_bin9
154 prop_binSR_electron2018_bin17_WpWpJJ_QCD
155 prop_binCROS_muon2017_bin4_Fake
156 prop_binSR_muon2017_bin15_sm
157 prop_binSR_muon2016M_bin15_TTTo2L2Nu
158 prop_binCRF_muon2018_bin13_sm
159 prop_binSR_electron2018_bin17_TVX
160 jer_2017
161 prop_binSR_muon2018_bin0
162 prop_binCRTT_muon2017_bin6
163 prop_binSR_muon2017_bin0
164 prop_binSR_muon2016M_bin12
165 prop_binSR_muon2017_bin12
166 prop_binCROS_muon2018_bin14
167 prop_binCROS_electron2017_bin4
168 prop_binCROS_electron2017_bin8
169 prop_binSR_muon2017_bin10_Fake
170 prop_binCRTT_muon2017_bin10
171 prop_binCRTT_muon2016M_bin3
172 prop_binCROS_electron2016M_bin1
173 prop_binSR_muon2018_bin10
174 prop_binSR_electron2018_bin11
175 prop_binCRTT_muon2017_bin1
176 prop_binCROS_electron2018_bin12_Fake
177 prop_binCRTT_muon2017_bin5
178 prop_binSR_electron2018_bin3
179 prop_binCRTT_electron2018_bin1
180 prop_binSR_electron2016M_bin12
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



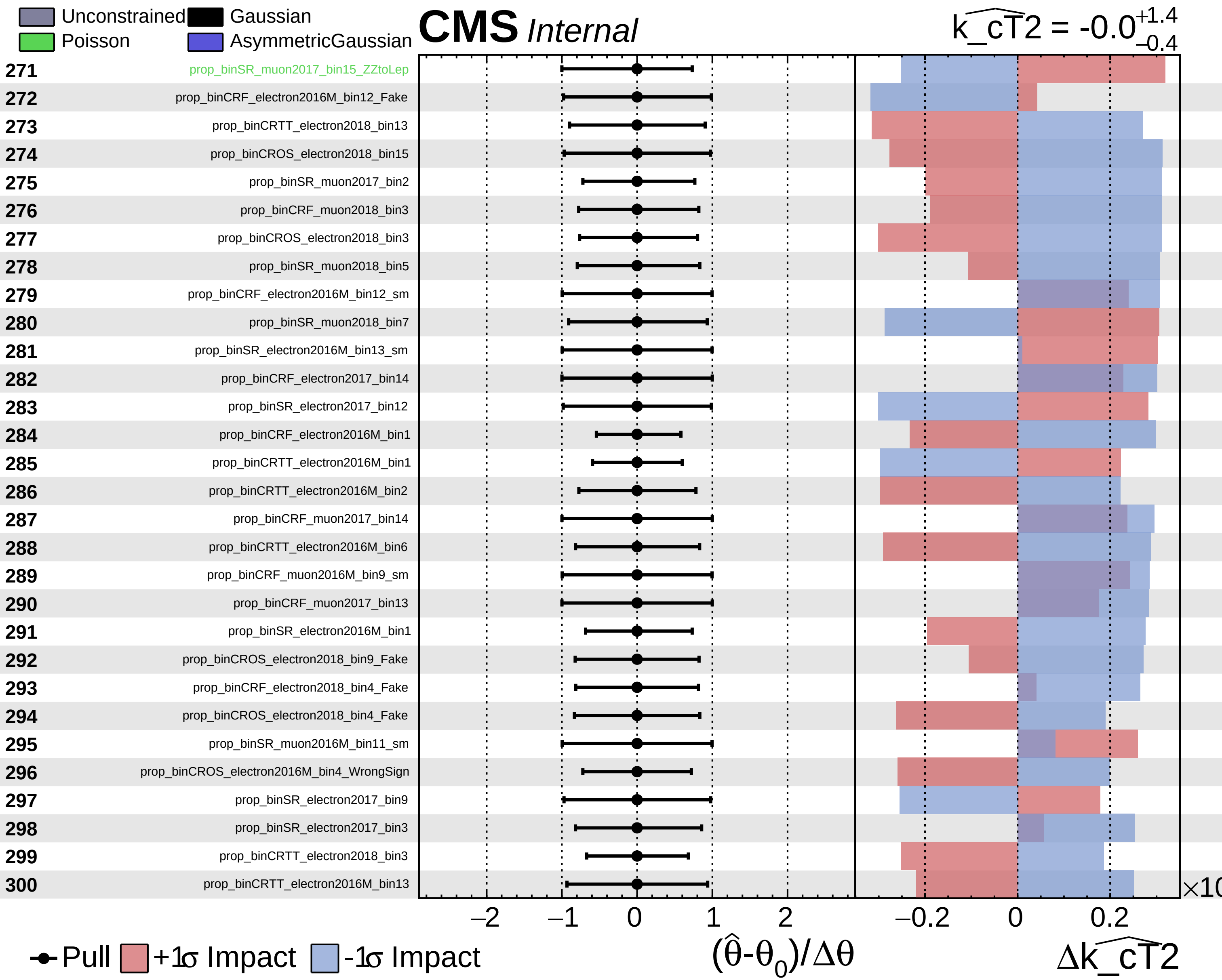
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+1.4}_{-0.4}$
181 prop_binCROS_electron2018_bin8
182 prop_binCROS_electron2016M_bin8_Fake
183 prop_binSR_muon2017_bin11
184 prop_binCRTT_muon2016M_bin2
185 prop_binCROS_muon2016M_bin10_WrongSign
186 prop_binCRTT_muon2018_bin14
187 prop_binCRTT_electron2016M_bin5
188 prop_binCRTT_electron2018_bin6
189 prop_binCROS_muon2016M_bin6
190 prop_binSR_muon2016M_bin13_WZ
191 prop_binCROS_muon2017_bin1
192 prop_binCRTT_muon2018_bin3
193 prop_binSR_muon2016M_bin9
194 prop_binCROS_muon2018_bin5_Fake
195 prop_binSR_muon2018_bin1
196 prop_binCRTT_muon2016M_bin9
197 prop_binSR_muon2016M_bin14_WrongSign
198 prop_binCRTT_electron2018_bin5
199 prop_binCRTT_muon2018_bin11
200 prop_binCRTT_muon2016M_bin8
201 prop_binCROS_muon2017_bin7_Fake
202 prop_binCRTT_electron2018_bin7
203 prop_binSR_electron2018_bin12_sm
204 metUnclust
205 prop_binCRTT_electron2017_bin2
206 prop_binCRTT_muon2017_bin9
207 prop_binCRTT_muon2018_bin8
208 prop_binCROS_electron2016M_bin3_WrongSign
209 prop_binSR_muon2018_bin9
210 prop_binSR_muon2016M_bin14_sm
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



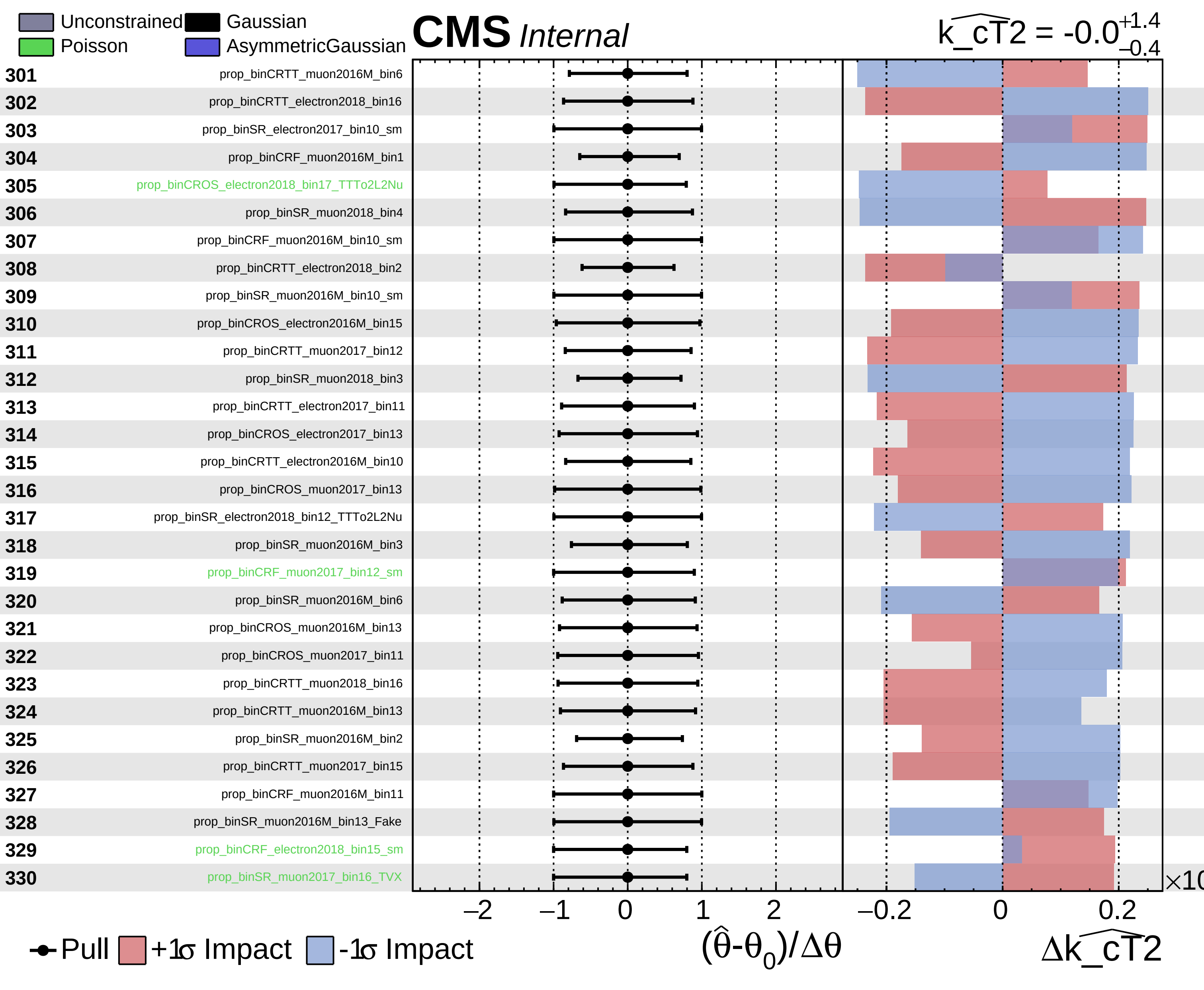
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +1.4 −0.4
211 puID
212 prop_binCROS_electron2018_bin4_WrongSign
213 prop_binSR_electron2016M_bin17_Triboson
214 prop_binCROS_muon2018_bin13
215 prop_binSR_electron2017_bin4
216 prop_binCROS_electron2016M_bin17_WrongSign
217 prop_binCRTT_muon2018_bin12
218 prop_binCRF_muon2018_bin15
219 prop_binCRTT_electron2016M_bin3
220 prop_binSR_electron2018_bin10_sm
221 prop_binCROS_electron2018_bin2
222 prop_binSR_electron2018_bin9
223 prop_binSR_muon2018_bin6
224 prop_binCRTT_muon2017_bin3
225 prop_binCRTT_muon2017_bin11
226 prop_binSR_muon2018_bin2
227 prop_binCRF_electron2018_bin1
228 prop_binCRF_electron2018_bin15_TTTo2L2Nu
229 prop_binSR_muon2017_bin13
230 prop_binCROS_muon2018_bin12
231 prop_binCROS_muon2017_bin12
232 prop_binCRF_muon2018_bin1
233 prop_binCROS_muon2018_bin16_WrongSign
234 prop_binSR_electron2017_bin13_sm
235 prop_binCRTT_electron2017_bin5
236 prop_binCROS_electron2018_bin11
237 prop_binCROS_muon2018_bin9_Fake
238 prop_binCRTT_muon2018_bin15
239 prop_binSR_muon2017_bin9
240 prop_binCROS_electron2017_bin7_Fake
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2



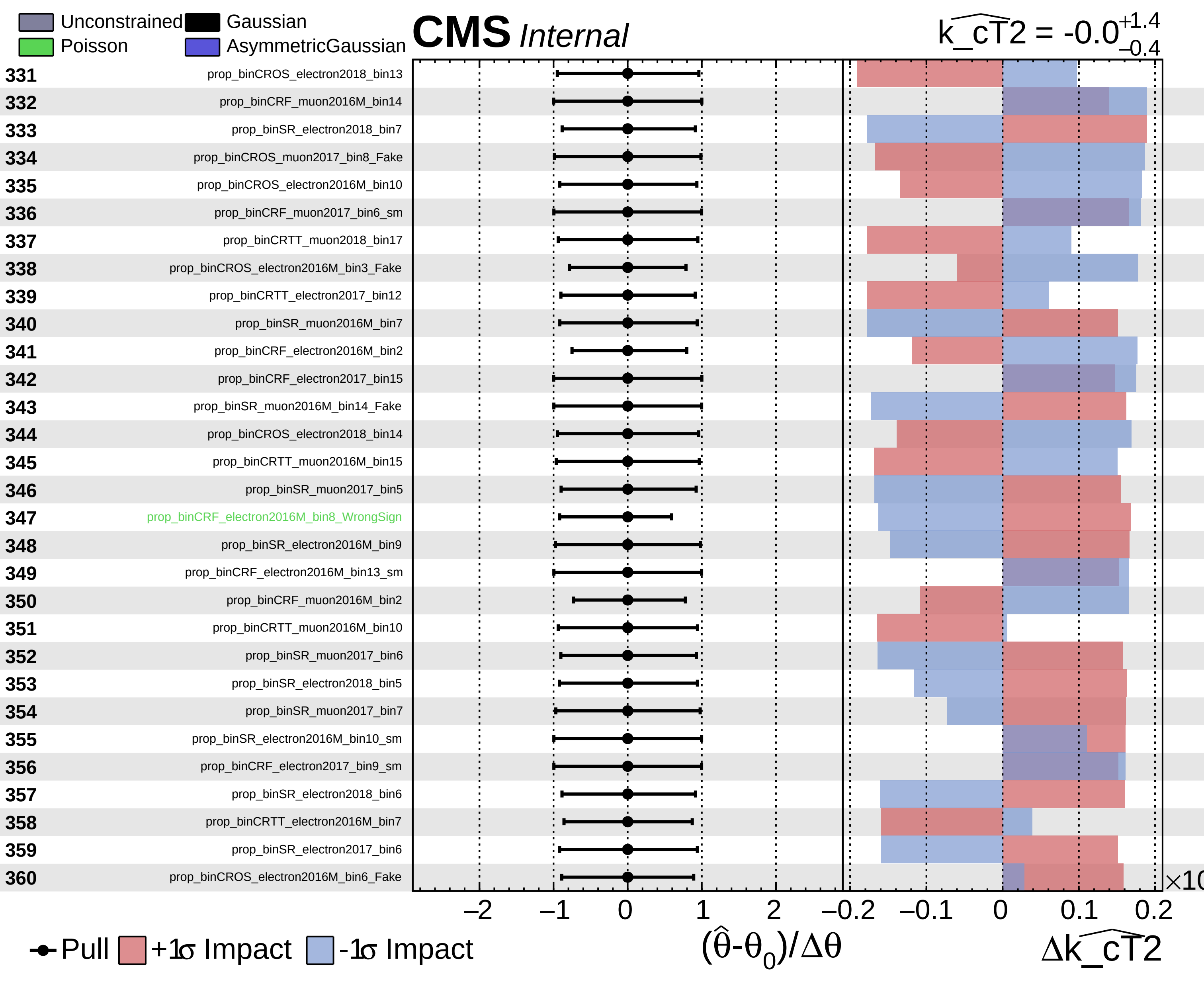
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+1.4}_{-0.4}$
241 prop_binCRF_electron2016M_bin11_sm
242 prop_binCROS_muon2016M_bin11_WrongSign
243 prop_binSR_muon2017_bin3
244 prop_binSR_muon2016M_bin14_TTTo2L2Nu
245 prop_binSR_muon2016M_bin1
246 prop_binSR_muon2016M_bin13_sm
247 prop_binSR_electron2018_bin4
248 prop_binCRTT_muon2018_bin13
249 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
250 prop_binSR_muon2016M_bin0
251 prop_binCROS_muon2018_bin10_Fake
252 prop_binCRF_electron2018_bin4_WrongSign
253 prop_binCRF_muon2018_bin2
254 prop_binSR_electron2018_bin8
255 prop_binSR_electron2016M_bin0
256 prop_binCRTT_electron2018_bin12
257 prop_binCRTT_electron2018_bin11
258 prop_binCRTT_electron2016M_bin12
259 prop_binCRF_electron2018_bin2
260 prop_binSR_electron2016M_bin13_TTTo2L2Nu
261 prop_binCRF_muon2017_bin7_sm
262 prop_binSR_electron2016M_bin11
263 prop_binSR_muon2017_bin1
264 prop_binCRF_muon2017_bin1
265 prop_binCRTT_muon2017_bin8
266 prop_binSR_electron2018_bin2
267 prop_binSR_electron2017_bin1
268 prop_binSR_electron2017_bin11_sm
269 prop_binSR_electron2016M_bin13_WZ
270 prop_binCRTT_muon2016M_bin12
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



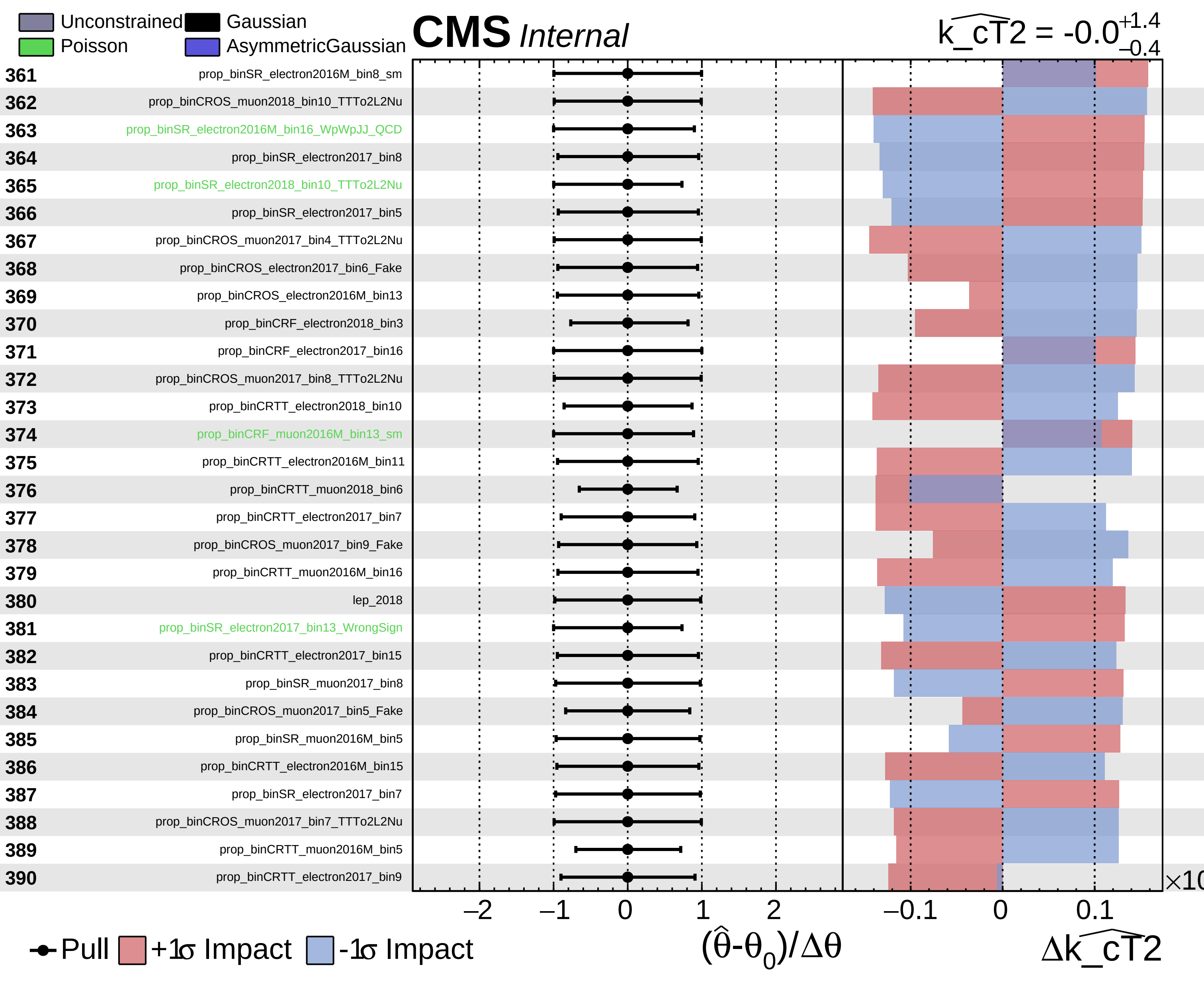
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+1.4}_{-0.4}$
271 prop_binSR_muon2017_bin15_ZZtoLep
272 prop_binCRF_electron2016M_bin12_Fake
273 prop_binCRTT_electron2018_bin13
274 prop_binCROS_electron2018_bin15
275 prop_binSR_muon2017_bin2
276 prop_binCRF_muon2018_bin3
277 prop_binCROS_electron2018_bin3
278 prop_binSR_muon2018_bin5
279 prop_binCRF_electron2016M_bin12_sm
280 prop_binSR_muon2018_bin7
281 prop_binSR_electron2016M_bin13_sm
282 prop_binCRF_electron2017_bin14
283 prop_binSR_electron2017_bin12
284 prop_binCRF_electron2016M_bin1
285 prop_binCRTT_electron2016M_bin1
286 prop_binCRTT_electron2016M_bin2
287 prop_binCRF_muon2017_bin14
288 prop_binCRTT_electron2016M_bin6
289 prop_binCRF_muon2016M_bin9_sm
290 prop_binCRF_muon2017_bin13
291 prop_binSR_electron2016M_bin1
292 prop_binCROS_electron2018_bin9_Fake
293 prop_binCRF_electron2018_bin4_Fake
294 prop_binCROS_electron2018_bin4_Fake
295 prop_binSR_muon2016M_bin11_sm
296 prop_binCROS_electron2016M_bin4_WrongSign
297 prop_binSR_electron2017_bin9
298 prop_binSR_electron2017_bin3
299 prop_binCRTT_electron2018_bin3
300 prop_binCRTT_electron2016M_bin13
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



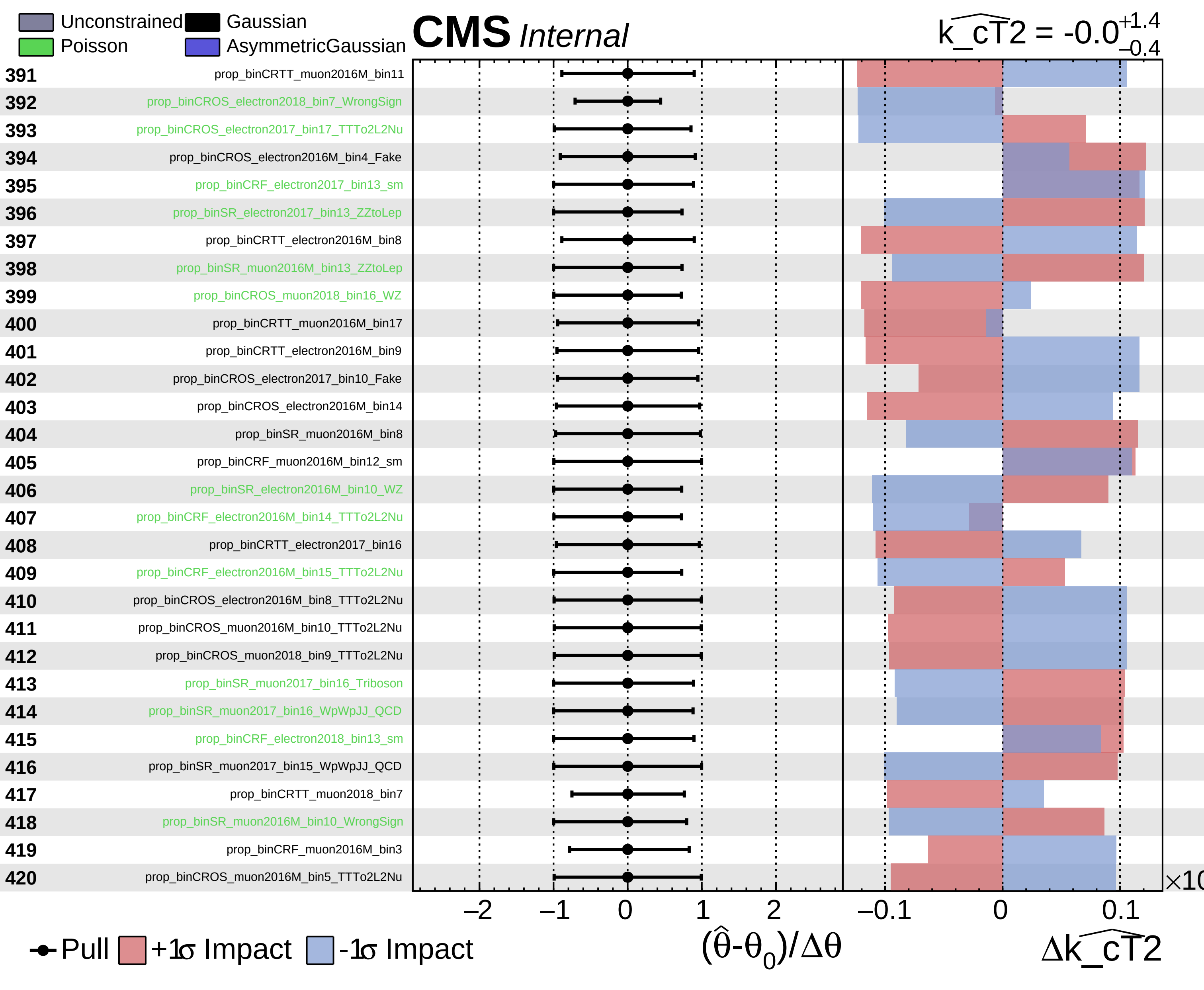
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+1.4}_{-0.4}$
301 prop_binCRTT_muon2016M_bin6
302 prop_binCRTT_electron2018_bin16
303 prop_binSR_electron2017_bin10_sm
304 prop_binCRF_muon2016M_bin1
305 prop_binCROS_electron2018_bin17_TTTo2L2Nu
306 prop_binSR_muon2018_bin4
307 prop_binCRF_muon2016M_bin10_sm
308 prop_binCRTT_electron2018_bin2
309 prop_binSR_muon2016M_bin10_sm
310 prop_binCROS_electron2016M_bin15
311 prop_binCRTT_muon2017_bin12
312 prop_binSR_muon2018_bin3
313 prop_binCRTT_electron2017_bin11
314 prop_binCROS_electron2017_bin13
315 prop_binCRTT_electron2016M_bin10
316 prop_binCROS_muon2017_bin13
317 prop_binSR_electron2018_bin12_TTTo2L2Nu
318 prop_binSR_muon2016M_bin3
319 prop_binCRF_muon2017_bin12_sm
320 prop_binSR_muon2016M_bin6
321 prop_binCROS_muon2016M_bin13
322 prop_binCROS_muon2017_bin11
323 prop_binCRTT_muon2018_bin16
324 prop_binCRTT_muon2016M_bin13
325 prop_binSR_muon2016M_bin2
326 prop_binCRTT_muon2017_bin15
327 prop_binCRF_muon2016M_bin11
328 prop_binSR_muon2016M_bin13_Fake
329 prop_binCRF_electron2018_bin15_sm
330 prop_binSR_muon2017_bin16_TVX
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



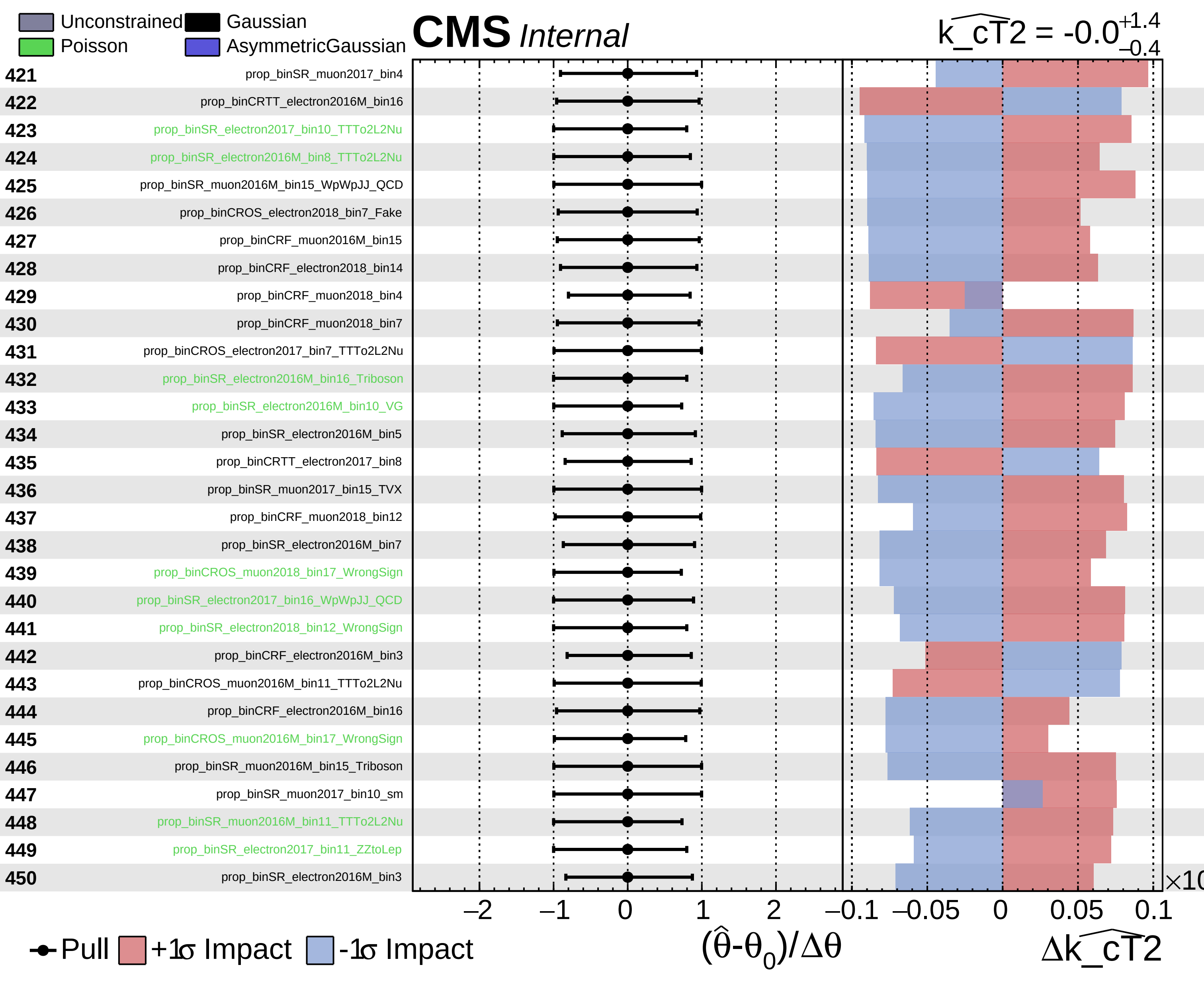
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +1.4 −0.4
331 prop_binCROS_electron2018_bin13
332 prop_binCRF_muon2016M_bin14
333 prop_binSR_electron2018_bin7
334 prop_binCROS_muon2017_bin8_Fake
335 prop_binCROS_electron2016M_bin10
336 prop_binCRF_muon2017_bin6_sm
337 prop_binCRTT_muon2018_bin17
338 prop_binCROS_electron2016M_bin3_Fake
339 prop_binCRTT_electron2017_bin12
340 prop_binSR_muon2016M_bin7
341 prop_binCRF_electron2016M_bin2
342 prop_binCRF_electron2017_bin15
343 prop_binSR_muon2016M_bin14_Fake
344 prop_binCROS_electron2018_bin14
345 prop_binCRTT_muon2016M_bin15
346 prop_binSR_muon2017_bin5
347 prop_binCRF_electron2016M_bin8_WrongSign
348 prop_binSR_electron2016M_bin9
349 prop_binCRF_electron2016M_bin13_sm
350 prop_binCRF_muon2016M_bin2
351 prop_binCRTT_muon2016M_bin10
352 prop_binSR_muon2017_bin6
353 prop_binSR_electron2018_bin5
354 prop_binSR_muon2017_bin7
355 prop_binSR_electron2016M_bin10_sm
356 prop_binCRF_electron2017_bin9_sm
357 prop_binSR_electron2018_bin6
358 prop_binCRTT_electron2016M_bin7
359 prop_binSR_electron2017_bin6
360 prop_binCROS_electron2016M_bin6_Fake
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



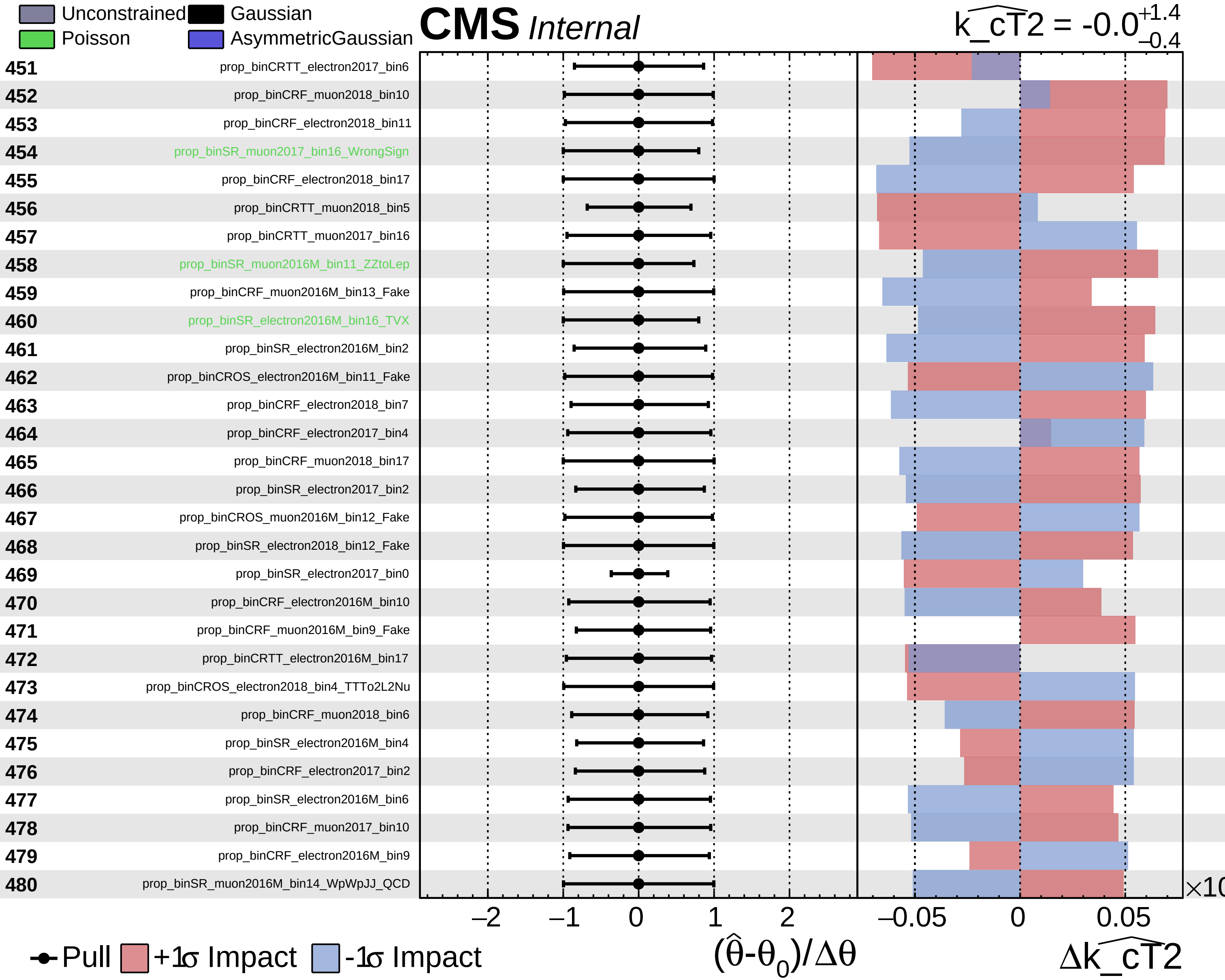
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +1.4 −0.4
361 prop_binSR_electron2016M_bin8_sm
362 prop_binCROS_muon2018_bin10_TTTo2L2Nu
363 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
364 prop_binSR_electron2017_bin8
365 prop_binSR_electron2018_bin10_TTTo2L2Nu
366 prop_binSR_electron2017_bin5
367 prop_binCROS_muon2017_bin4_TTTo2L2Nu
368 prop_binCROS_electron2017_bin6_Fake
369 prop_binCROS_electron2016M_bin13
370 prop_binCRF_electron2018_bin3
371 prop_binCRF_electron2017_bin16
372 prop_binCROS_muon2017_bin8_TTTo2L2Nu
373 prop_binCRTT_electron2018_bin10
374 prop_binCRF_muon2016M_bin13_sm
375 prop_binCRTT_electron2016M_bin11
376 prop_binCRTT_muon2018_bin6
377 prop_binCRTT_electron2017_bin7
378 prop_binCROS_muon2017_bin9_Fake
379 prop_binCRTT_muon2016M_bin16
380 lep_2018
381 prop_binSR_electron2017_bin13_WrongSign
382 prop_binCRTT_electron2017_bin15
383 prop_binSR_muon2017_bin8
384 prop_binCROS_muon2017_bin5_Fake
385 prop_binSR_muon2016M_bin5
386 prop_binCRTT_electron2016M_bin15
387 prop_binSR_electron2017_bin7
388 prop_binCROS_muon2017_bin7_TTTo2L2Nu
389 prop_binCRTT_muon2016M_bin5
390 prop_binCRTT_electron2017_bin9
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

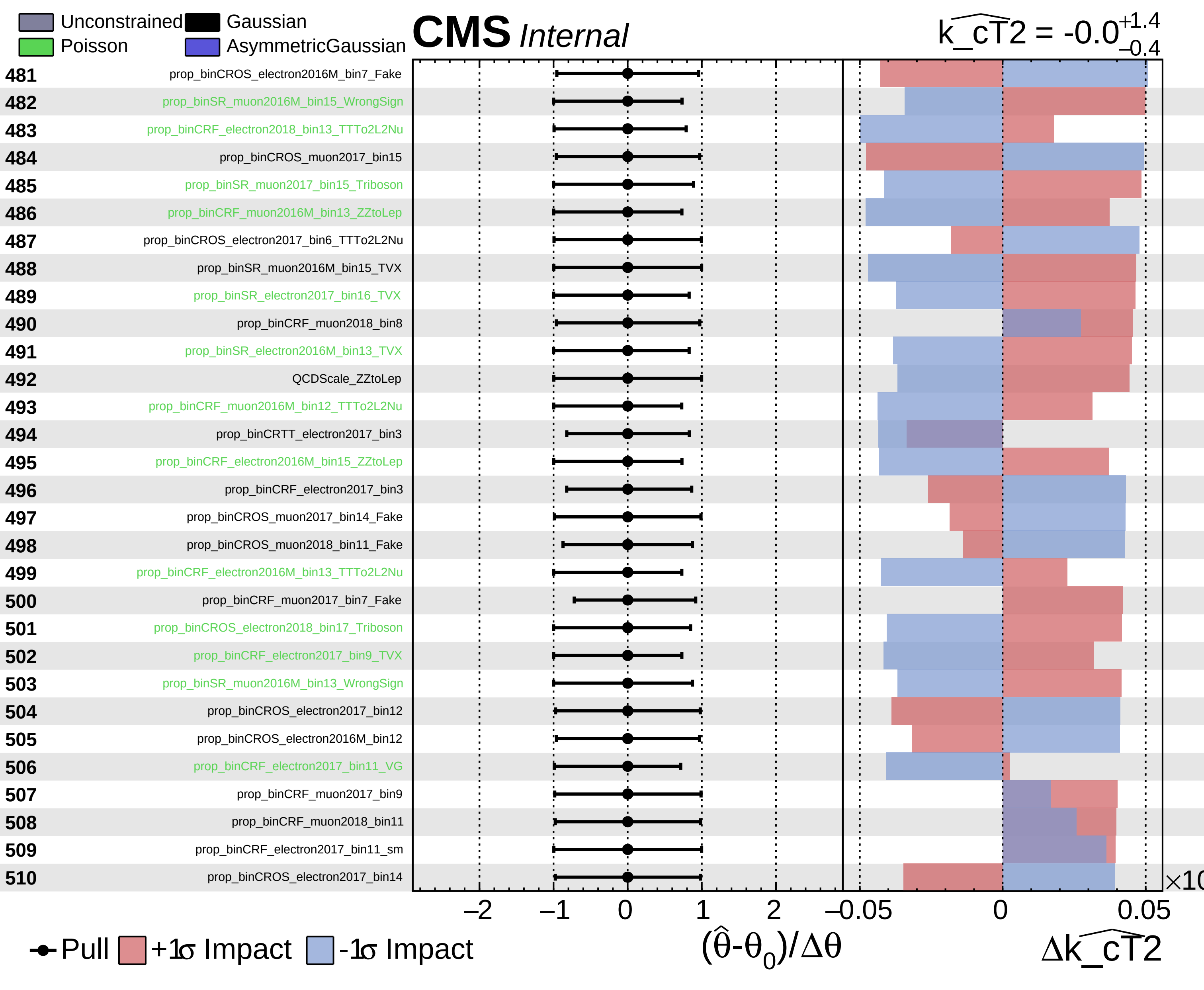


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +1.4 −0.4
391 prop_binCRTT_muon2016M_bin11
392 prop_binCROS_electron2018_bin7_WrongSign
393 prop_binCROS_electron2017_bin17_TTTo2L2Nu
394 prop_binCROS_electron2016M_bin4_Fake
395 prop_binCRF_electron2017_bin13_sm
396 prop_binSR_electron2017_bin13_ZZtoLep
397 prop_binCRTT_electron2016M_bin8
398 prop_binSR_muon2016M_bin13_ZZtoLep
399 prop_binCROS_muon2018_bin16_WZ
400 prop_binCRTT_muon2016M_bin17
401 prop_binCRTT_electron2016M_bin9
402 prop_binCROS_electron2017_bin10_Fake
403 prop_binCROS_electron2016M_bin14
404 prop_binSR_muon2016M_bin8
405 prop_binCRF_muon2016M_bin12_sm
406 prop_binSR_electron2016M_bin10_WZ
407 prop_binCRF_electron2016M_bin14_TTTo2L2Nu
408 prop_binCRTT_electron2017_bin16
409 prop_binCRF_electron2016M_bin15_TTTo2L2Nu
410 prop_binCROS_electron2016M_bin8_TTTo2L2Nu
411 prop_binCROS_muon2016M_bin10_TTTo2L2Nu
412 prop_binCROS_muon2018_bin9_TTTo2L2Nu
413 prop_binSR_muon2017_bin16_Triboson
414 prop_binSR_muon2017_bin16_WpWpJJ_QCD
415 prop_binCRF_electron2018_bin13_sm
416 prop_binSR_muon2017_bin15_WpWpJJ_QCD
417 prop_binCRTT_muon2018_bin7
418 prop_binSR_muon2016M_bin10_WrongSign
419 prop_binCRF_muon2016M_bin3
420 prop_binCROS_muon2016M_bin5_TTTo2L2Nu
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2

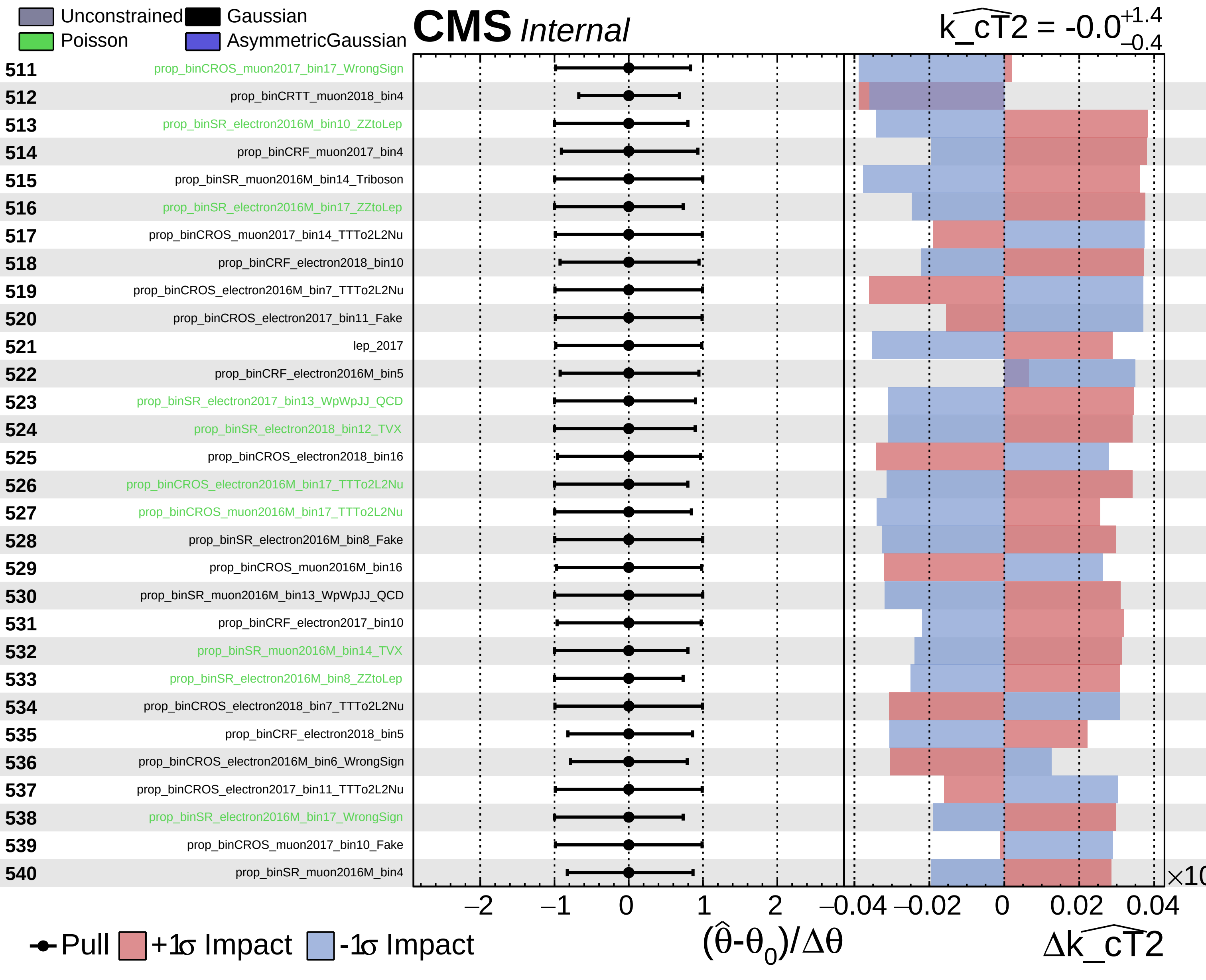


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +1.4 −0.4
421 prop_binSR_muon2017_bin4
422 prop_binCRTT_electron2016M_bin16
423 prop_binSR_electron2017_bin10_TTTo2L2Nu
424 prop_binSR_electron2016M_bin8_TTTo2L2Nu
425 prop_binSR_muon2016M_bin15_WpWpJJ_QCD
426 prop_binCROS_electron2018_bin7_Fake
427 prop_binCRF_muon2016M_bin15
428 prop_binCRF_electron2018_bin14
429 prop_binCRF_muon2018_bin4
430 prop_binCRF_muon2018_bin7
431 prop_binCROS_electron2017_bin7_TTTo2L2Nu
432 prop_binSR_electron2016M_bin16_Triboson
433 prop_binSR_electron2016M_bin10_VG
434 prop_binSR_electron2016M_bin5
435 prop_binCRTT_electron2017_bin8
436 prop_binSR_muon2017_bin15_TVX
437 prop_binCRF_muon2018_bin12
438 prop_binSR_electron2016M_bin7
439 prop_binCROS_muon2018_bin17_WrongSign
440 prop_binSR_electron2017_bin16_WpWpJJ_QCD
441 prop_binSR_electron2018_bin12_WrongSign
442 prop_binCRF_electron2016M_bin3
443 prop_binCROS_muon2016M_bin11_TTTo2L2Nu
444 prop_binCRF_electron2016M_bin16
445 prop_binCROS_muon2016M_bin17_WrongSign
446 prop_binSR_muon2016M_bin15_Triboson
447 prop_binSR_muon2017_bin10_sm
448 prop_binSR_muon2016M_bin11_TTTo2L2Nu
449 prop_binSR_electron2017_bin11_ZZtoLep
450 prop_binSR_electron2016M_bin3
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2

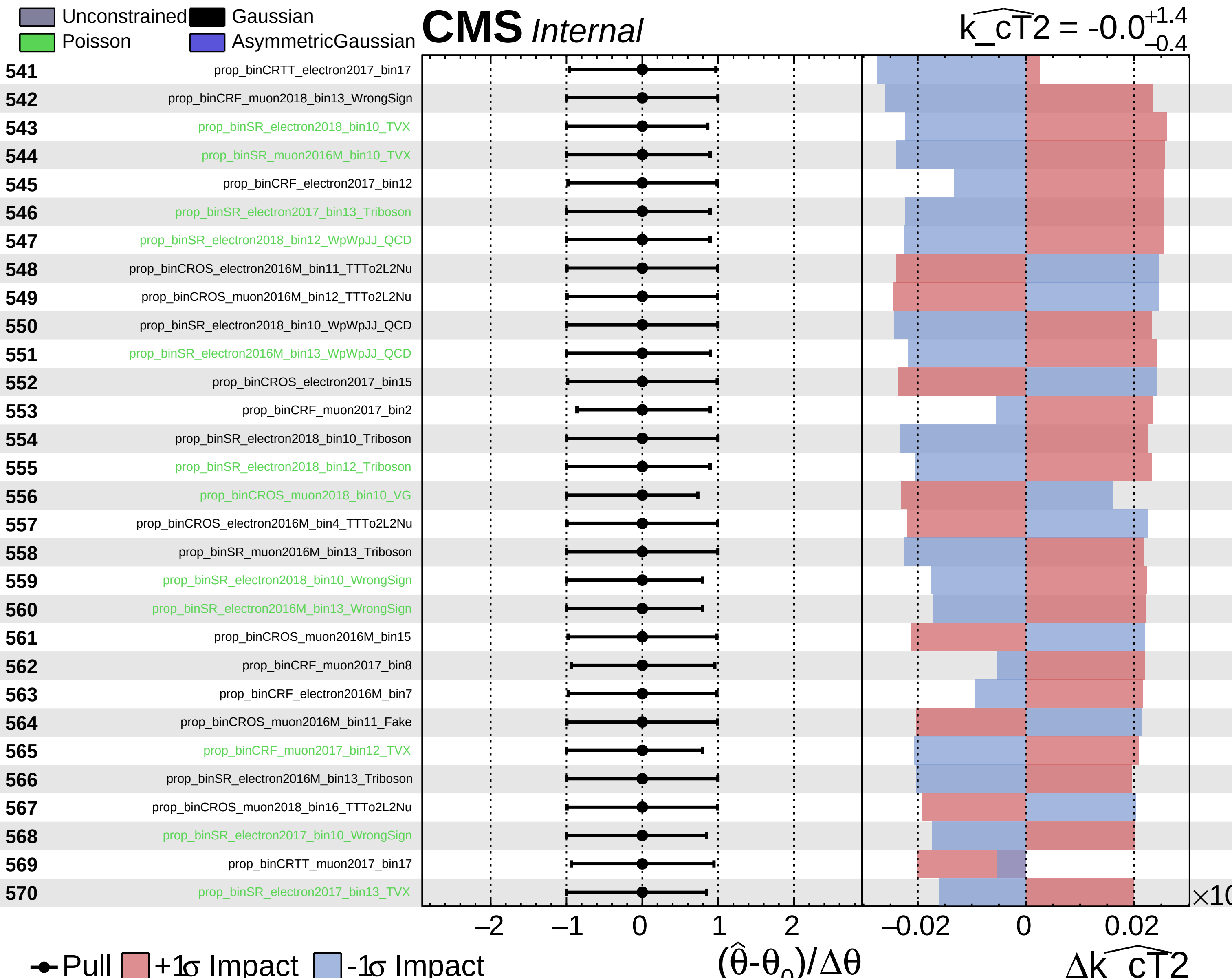

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +1.4 −0.4
451 prop_binCRTT_electron2017_bin6
452 prop_binCRF_muon2018_bin10
453 prop_binCRF_electron2018_bin11
454 prop_binSR_muon2017_bin16_WrongSign
455 prop_binCRF_electron2018_bin17
456 prop_binCRTT_muon2018_bin5
457 prop_binCRTT_muon2017_bin16
458 prop_binSR_muon2016M_bin11_ZZtoLep
459 prop_binCRF_muon2016M_bin13_Fake
460 prop_binSR_electron2016M_bin16_TVX
461 prop_binSR_electron2016M_bin2
462 prop_binCROS_electron2016M_bin11_Fake
463 prop_binCRF_electron2018_bin7
464 prop_binCRF_electron2017_bin4
465 prop_binCRF_muon2018_bin17
466 prop_binSR_electron2017_bin2
467 prop_binCROS_muon2016M_bin12_Fake
468 prop_binSR_electron2018_bin12_Fake
469 prop_binSR_electron2017_bin0
470 prop_binCRF_electron2016M_bin10
471 prop_binCRF_muon2016M_bin9_Fake
472 prop_binCRTT_electron2016M_bin17
473 prop_binCROS_electron2018_bin4_TTTo2L2Nu
474 prop_binCRF_muon2018_bin6
475 prop_binSR_electron2016M_bin4
476 prop_binCRF_electron2017_bin2
477 prop_binSR_electron2016M_bin6
478 prop_binCRF_muon2017_bin10
479 prop_binCRF_electron2016M_bin9
480 prop_binSR_muon2016M_bin14_WpWpJJ_QCD
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



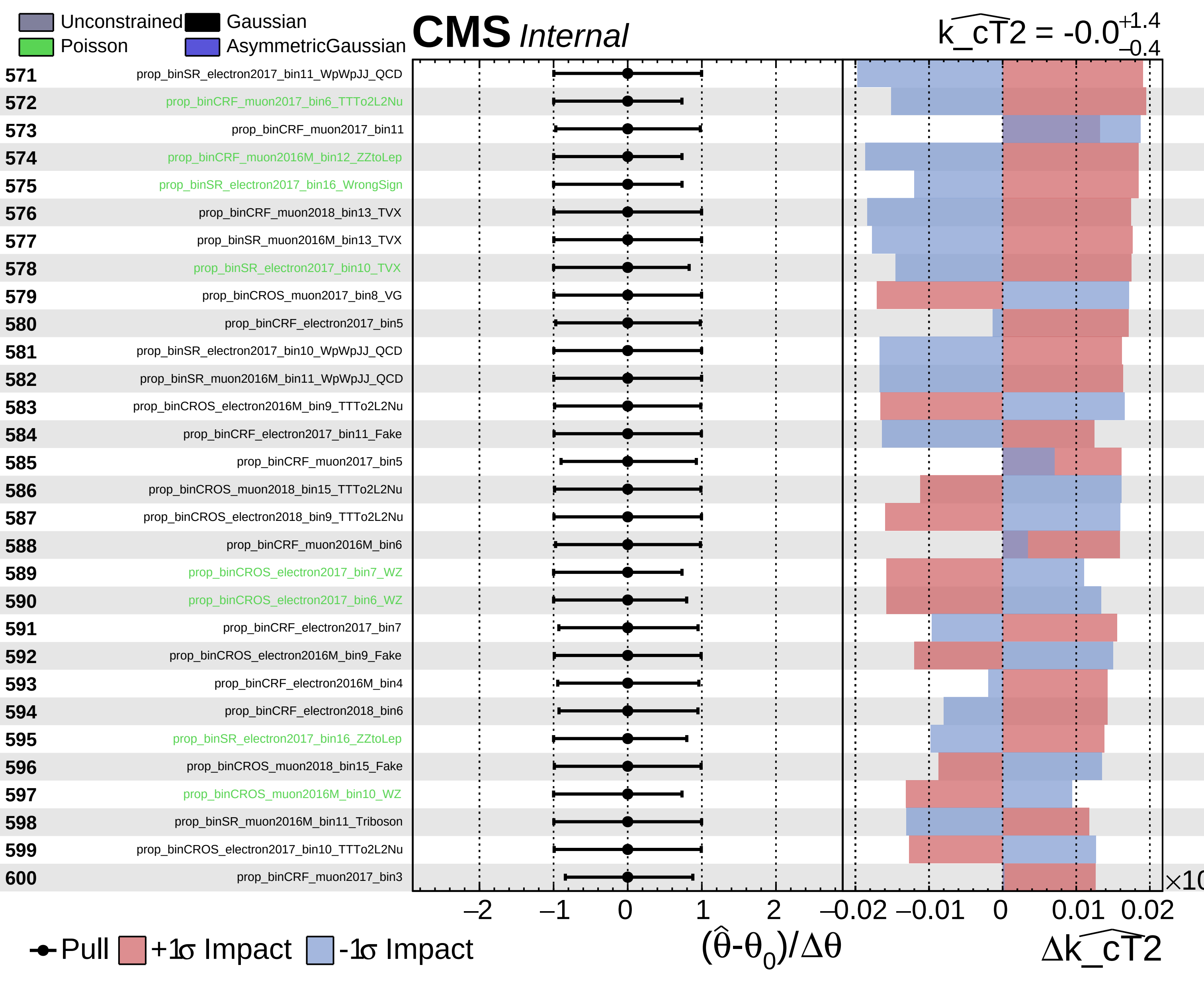
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+1.4}_{-0.4}$
481 prop_binCROS_electron2016M_bin7_Fake
482 prop_binSR_muon2016M_bin15_WrongSign
483 prop_binCRF_electron2018_bin13_TTTo2L2Nu
484 prop_binCROS_muon2017_bin15
485 prop_binSR_muon2017_bin15_Triboson
486 prop_binCRF_muon2016M_bin13_ZZtoLep
487 prop_binCROS_electron2017_bin6_TTTo2L2Nu
488 prop_binSR_muon2016M_bin15_TVX
489 prop_binSR_electron2017_bin16_TVX
490 prop_binCRF_muon2018_bin8
491 prop_binSR_electron2016M_bin13_TVX
492 QCDScale_ZZtoLep
493 prop_binCRF_muon2016M_bin12_TTTo2L2Nu
494 prop_binCRTT_electron2017_bin3
495 prop_binCRF_electron2016M_bin15_ZZtoLep
496 prop_binCRF_electron2017_bin3
497 prop_binCROS_muon2017_bin14_Fake
498 prop_binCROS_muon2018_bin11_Fake
499 prop_binCRF_electron2016M_bin13_TTTo2L2Nu
500 prop_binCRF_muon2017_bin7_Fake
501 prop_binCROS_electron2018_bin17_Triboson
502 prop_binCRF_electron2017_bin9_TVX
503 prop_binSR_muon2016M_bin13_WrongSign
504 prop_binCROS_electron2017_bin12
505 prop_binCROS_electron2016M_bin12
506 prop_binCRF_electron2017_bin11_VG
507 prop_binCRF_muon2017_bin9
508 prop_binCRF_muon2018_bin11
509 prop_binCRF_electron2017_bin11_sm
510 prop_binCROS_electron2017_bin14
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



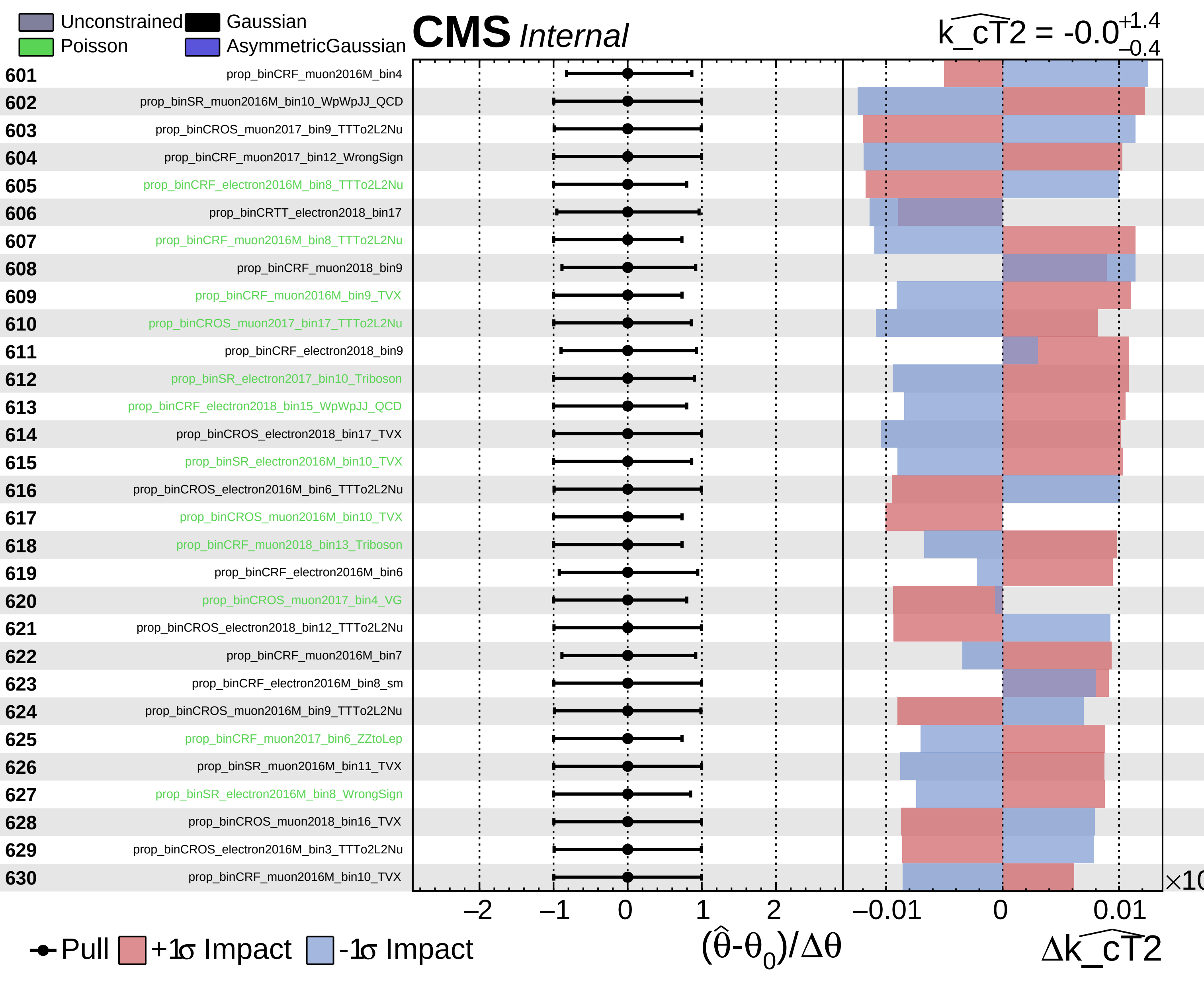
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +1.4 −0.4
511 prop_binCROS_muon2017_bin17_WrongSign
512 prop_binCRTT_muon2018_bin4
513 prop_binSR_electron2016M_bin10_ZZtoLep
514 prop_binCRF_muon2017_bin4
515 prop_binSR_muon2016M_bin14_Triboson
516 prop_binSR_electron2016M_bin17_ZZtoLep
517 prop_binCROS_muon2017_bin14_TTTo2L2Nu
518 prop_binCRF_electron2018_bin10
519 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
520 prop_binCROS_electron2017_bin11_Fake
521 lep_2017
522 prop_binCRF_electron2016M_bin5
523 prop_binSR_electron2017_bin13_WpWpJJ_QCD
524 prop_binSR_electron2018_bin12_TVX
525 prop_binCROS_electron2018_bin16
526 prop_binCROS_electron2016M_bin17_TTTo2L2Nu
527 prop_binCROS_muon2016M_bin17_TTTo2L2Nu
528 prop_binSR_electron2016M_bin8_Fake
529 prop_binCROS_muon2016M_bin16
530 prop_binSR_muon2016M_bin13_WpWpJJ_QCD
531 prop_binCRF_electron2017_bin10
532 prop_binSR_muon2016M_bin14_TVX
533 prop_binSR_electron2016M_bin8_ZZtoLep
534 prop_binCROS_electron2018_bin7_TTTo2L2Nu
535 prop_binCRF_electron2018_bin5
536 prop_binCROS_electron2016M_bin6_WrongSign
537 prop_binCROS_electron2017_bin11_TTTo2L2Nu
538 prop_binSR_electron2016M_bin17_WrongSign
539 prop_binCROS_muon2017_bin10_Fake
540 prop_binSR_muon2016M_bin4
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2

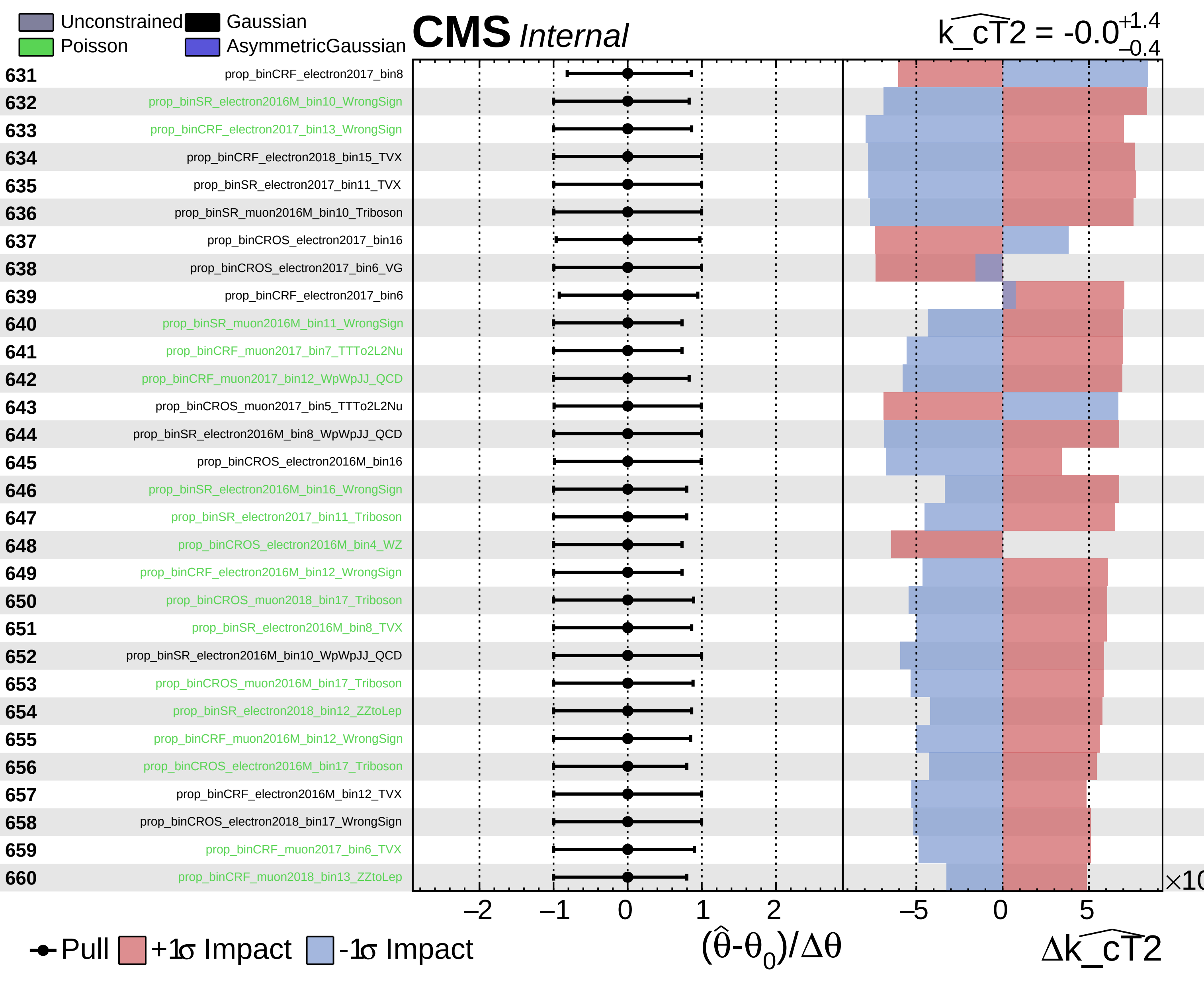


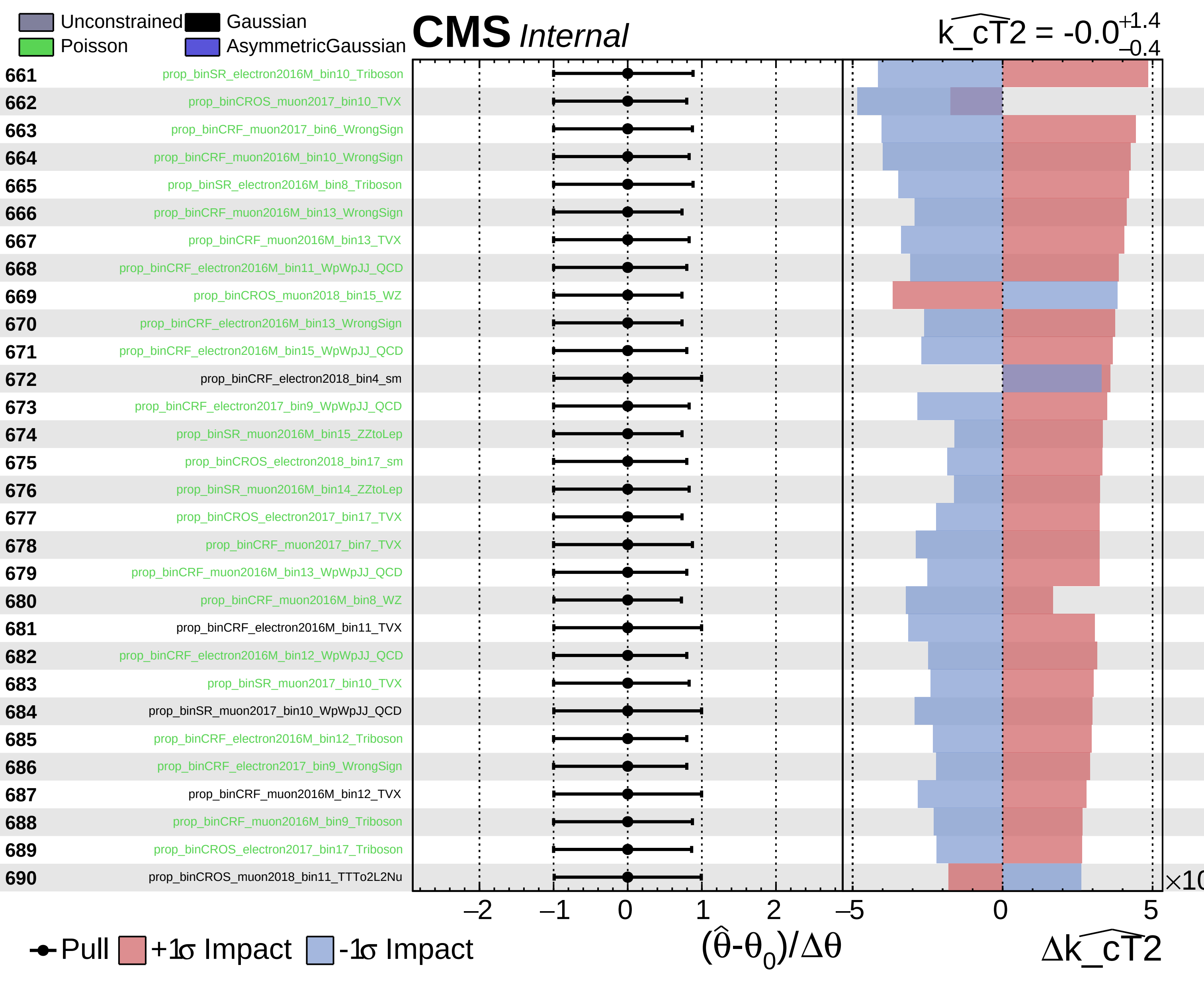
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+1.4}_{-0.4}$
541 prop_binCRTT_electron2017_bin17
542 prop_binCRF_muon2018_bin13_WrongSign
543 prop_binSR_electron2018_bin10_TVX
544 prop_binSR_muon2016M_bin10_TVX
545 prop_binCRF_electron2017_bin12
546 prop_binSR_electron2017_bin13_Triboson
547 prop_binSR_electron2018_bin12_WpWpJJ_QCD
548 prop_binCROS_electron2016M_bin11_TTTo2L2Nu
549 prop_binCROS_muon2016M_bin12_TTTo2L2Nu
550 prop_binSR_electron2018_bin10_WpWpJJ_QCD
551 prop_binSR_electron2016M_bin13_WpWpJJ_QCD
552 prop_binCROS_electron2017_bin15
553 prop_binCRF_muon2017_bin2
554 prop_binSR_electron2018_bin10_Triboson
555 prop_binSR_electron2018_bin12_Triboson
556 prop_binCROS_muon2018_bin10_VG
557 prop_binCROS_electron2016M_bin4_TTTo2L2Nu
558 prop_binSR_muon2016M_bin13_Triboson
559 prop_binSR_electron2018_bin10_WrongSign
560 prop_binSR_electron2016M_bin13_WrongSign
561 prop_binCROS_muon2016M_bin15
562 prop_binCRF_muon2017_bin8
563 prop_binCRF_electron2016M_bin7
564 prop_binCROS_muon2016M_bin11_Fake
565 prop_binCRF_muon2017_bin12_TVX
566 prop_binSR_electron2016M_bin13_Triboson
567 prop_binCROS_muon2018_bin16_TTTo2L2Nu
568 prop_binSR_electron2017_bin10_WrongSign
569 prop_binCRTT_muon2017_bin17
570 prop_binSR_electron2017_bin13_TVX
−2 −1 0 1 2
−0.02 0 0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

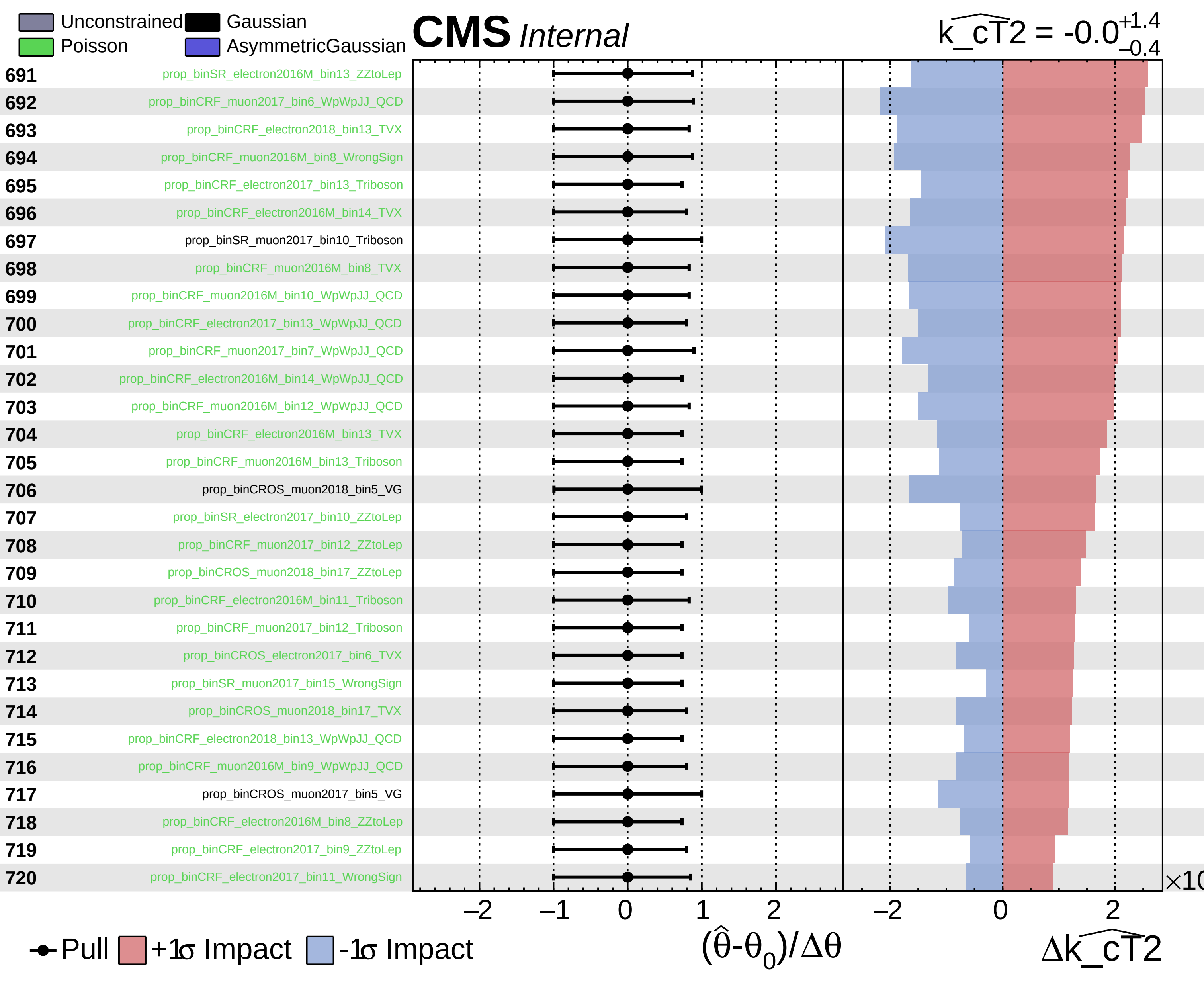


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +1.4 −0.4
571 prop_binSR_electron2017_bin11_WpWpJJ_QCD
572 prop_binCRF_muon2017_bin6_TTTo2L2Nu
573 prop_binCRF_muon2017_bin11
574 prop_binCRF_muon2016M_bin12_ZZtoLep
575 prop_binSR_electron2017_bin16_WrongSign
576 prop_binCRF_muon2018_bin13_TVX
577 prop_binSR_muon2016M_bin13_TVX
578 prop_binSR_electron2017_bin10_TVX
579 prop_binCROS_muon2017_bin8_VG
580 prop_binCRF_electron2017_bin5
581 prop_binSR_electron2017_bin10_WpWpJJ_QCD
582 prop_binSR_muon2016M_bin11_WpWpJJ_QCD
583 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
584 prop_binCRF_electron2017_bin11_Fake
585 prop_binCRF_muon2017_bin5
586 prop_binCROS_muon2018_bin15_TTTo2L2Nu
587 prop_binCROS_electron2018_bin9_TTTo2L2Nu
588 prop_binCRF_muon2016M_bin6
589 prop_binCROS_electron2017_bin7_WZ
590 prop_binCROS_electron2017_bin6_WZ
591 prop_binCRF_electron2017_bin7
592 prop_binCROS_electron2016M_bin9_Fake
593 prop_binCRF_electron2016M_bin4
594 prop_binCRF_electron2018_bin6
595 prop_binSR_electron2017_bin16_ZZtoLep
596 prop_binCROS_muon2018_bin15_Fake
597 prop_binCROS_muon2016M_bin10_WZ
598 prop_binSR_muon2016M_bin11_Triboson
599 prop_binCROS_electron2017_bin10_TTTo2L2Nu
600 prop_binCRF_muon2017_bin3
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

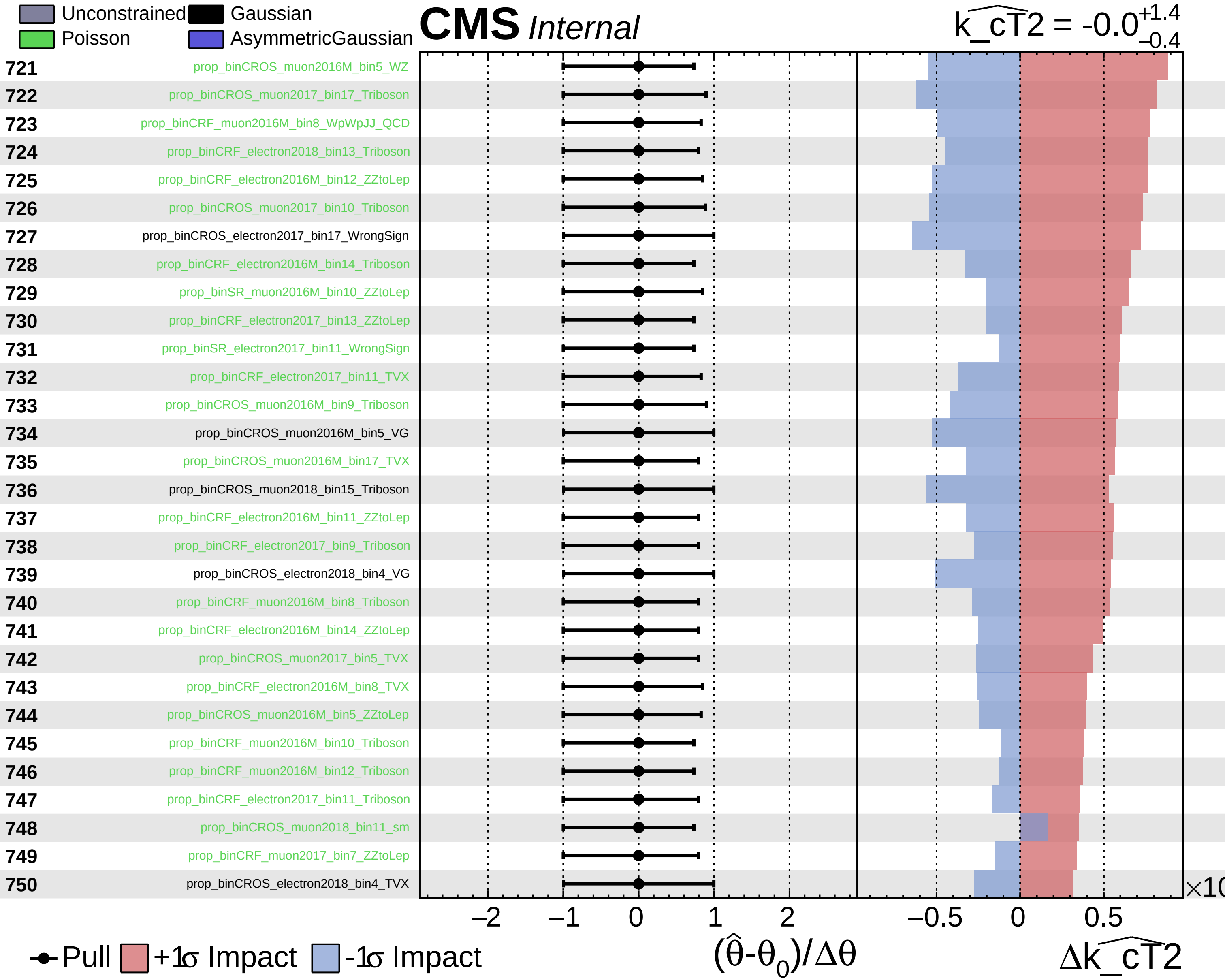

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0+1.4−0.4
601 prop_binCRF_muon2016M_bin4
602 prop_binSR_muon2016M_bin10_WpWpJJ_QCD
603 prop_binCROS_muon2017_bin9_TTTo2L2Nu
604 prop_binCRF_muon2017_bin12_WrongSign
605 prop_binCRF_electron2016M_bin8_TTTo2L2Nu
606 prop_binCRTT_electron2018_bin17
607 prop_binCRF_muon2016M_bin8_TTTo2L2Nu
608 prop_binCRF_muon2018_bin9
609 prop_binCRF_muon2016M_bin9_TVX
610 prop_binCROS_muon2017_bin17_TTTo2L2Nu
611 prop_binCRF_electron2018_bin9
612 prop_binSR_electron2017_bin10_Triboson
613 prop_binCRF_electron2018_bin15_WpWpJJ_QCD
614 prop_binCROS_electron2018_bin17_TVX
615 prop_binSR_electron2016M_bin10_TVX
616 prop_binCROS_electron2016M_bin6_TTTo2L2Nu
617 prop_binCROS_muon2016M_bin10_TVX
618 prop_binCRF_muon2018_bin13_Triboson
619 prop_binCRF_electron2016M_bin6
620 prop_binCROS_muon2017_bin4_VG
621 prop_binCROS_electron2018_bin12_TTTo2L2Nu
622 prop_binCRF_muon2016M_bin7
623 prop_binCRF_electron2016M_bin8_sm
624 prop_binCROS_muon2016M_bin9_TTTo2L2Nu
625 prop_binCRF_muon2017_bin6_ZZtoLep
626 prop_binSR_muon2016M_bin11_TVX
627 prop_binSR_electron2016M_bin8_WrongSign
628 prop_binCROS_muon2018_bin16_TVX
629 prop_binCROS_electron2016M_bin3_TTTo2L2Nu
630 prop_binCRF_muon2016M_bin10_TVX
−2 −1 0 1 2
−0.01 0 0.01
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+1.4}_{-0.4}$
631 prop_binCRF_electron2017_bin8
632 prop_binSR_electron2016M_bin10_WrongSign
633 prop_binCRF_electron2017_bin13_WrongSign
634 prop_binCRF_electron2018_bin15_TVX
635 prop_binSR_electron2017_bin11_TVX
636 prop_binSR_muon2016M_bin10_Triboson
637 prop_binCROS_electron2017_bin16
638 prop_binCROS_electron2017_bin6_VG
639 prop_binCRF_electron2017_bin6
640 prop_binSR_muon2016M_bin11_WrongSign
641 prop_binCRF_muon2017_bin7_TTTo2L2Nu
642 prop_binCRF_muon2017_bin12_WpWpJJ_QCD
643 prop_binCROS_muon2017_bin5_TTTo2L2Nu
644 prop_binSR_electron2016M_bin8_WpWpJJ_QCD
645 prop_binCROS_electron2016M_bin16
646 prop_binSR_electron2016M_bin16_WrongSign
647 prop_binSR_electron2017_bin11_Triboson
648 prop_binCROS_electron2016M_bin4_WZ
649 prop_binCRF_electron2016M_bin12_WrongSign
650 prop_binCROS_muon2018_bin17_Triboson
651 prop_binSR_electron2016M_bin8_TVX
652 prop_binSR_electron2016M_bin10_WpWpJJ_QCD
653 prop_binCROS_muon2016M_bin17_Triboson
654 prop_binSR_electron2018_bin12_ZZtoLep
655 prop_binCRF_muon2016M_bin12_WrongSign
656 prop_binCROS_electron2016M_bin17_Triboson
657 prop_binCRF_electron2016M_bin12_TVX
658 prop_binCROS_electron2018_bin17_WrongSign
659 prop_binCRF_muon2017_bin6_TVX
660 prop_binCRF_muon2018_bin13_ZZtoLep
−2 −1 0 1 2
−5 0 5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

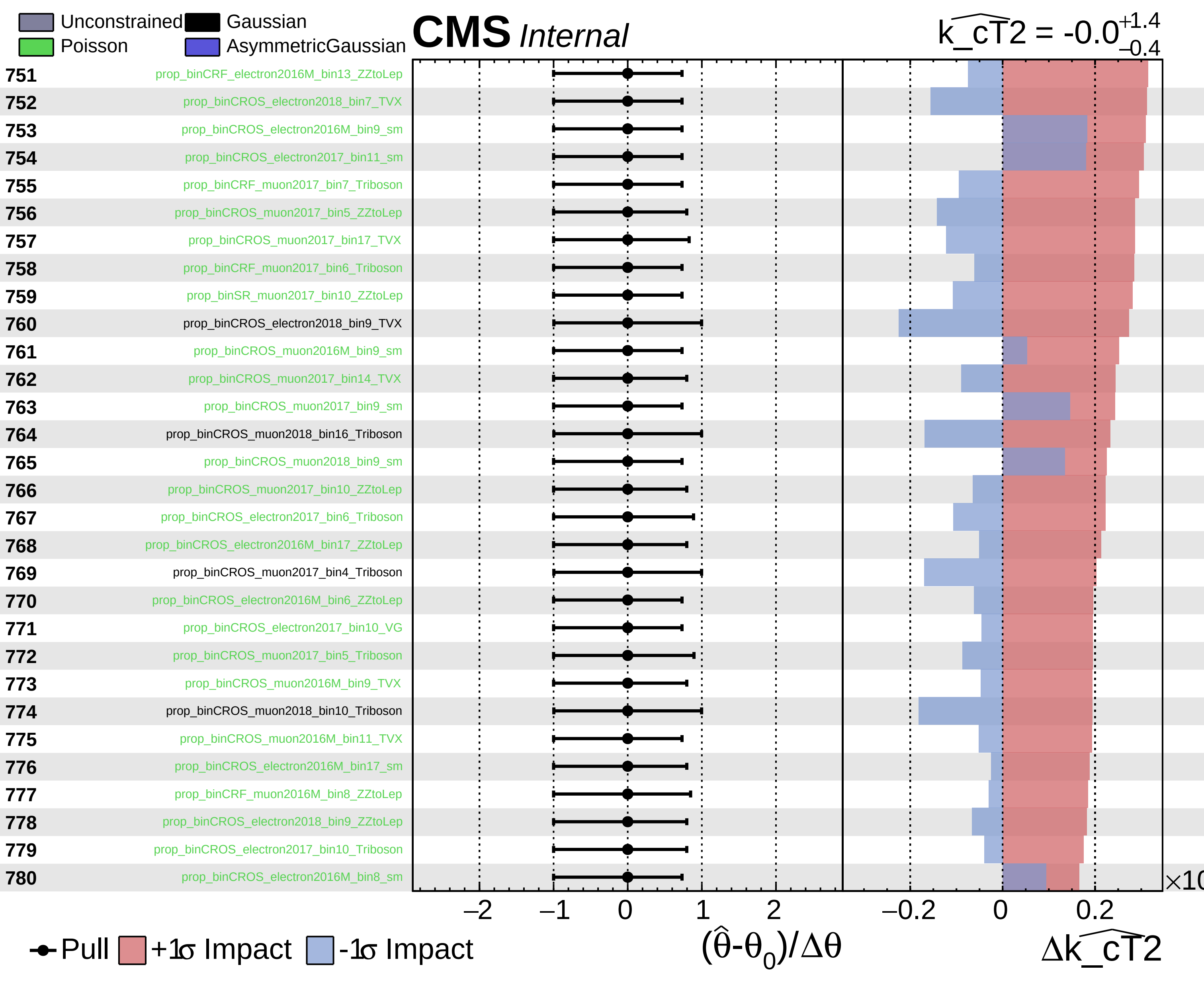

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +1.4 −0.4
661 prop_binSR_electron2016M_bin10_Triboson
662 prop_binCROS_muon2017_bin10_TVX
663 prop_binCRF_muon2017_bin6_WrongSign
664 prop_binCRF_muon2016M_bin10_WrongSign
665 prop_binSR_electron2016M_bin8_Triboson
666 prop_binCRF_muon2016M_bin13_WrongSign
667 prop_binCRF_muon2016M_bin13_TVX
668 prop_binCRF_electron2016M_bin11_WpWpJJ_QCD
669 prop_binCROS_muon2018_bin15_WZ
670 prop_binCRF_electron2016M_bin13_WrongSign
671 prop_binCRF_electron2016M_bin15_WpWpJJ_QCD
672 prop_binCRF_electron2018_bin4_sm
673 prop_binCRF_electron2017_bin9_WpWpJJ_QCD
674 prop_binSR_muon2016M_bin15_ZZtoLep
675 prop_binCROS_electron2018_bin17_sm
676 prop_binSR_muon2016M_bin14_ZZtoLep
677 prop_binCROS_electron2017_bin17_TVX
678 prop_binCRF_muon2017_bin7_TVX
679 prop_binCRF_muon2016M_bin13_WpWpJJ_QCD
680 prop_binCRF_muon2016M_bin8_WZ
681 prop_binCRF_electron2016M_bin11_TVX
682 prop_binCRF_electron2016M_bin12_WpWpJJ_QCD
683 prop_binSR_muon2017_bin10_TVX
684 prop_binSR_muon2017_bin10_WpWpJJ_QCD
685 prop_binCRF_electron2016M_bin12_Triboson
686 prop_binCRF_electron2017_bin9_WrongSign
687 prop_binCRF_muon2016M_bin12_TVX
688 prop_binCRF_muon2016M_bin9_Triboson
689 prop_binCROS_electron2017_bin17_Triboson
690 prop_binCROS_muon2018_bin11_TTTo2L2Nu
−2 −1 0 1 2
−5 0 5
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



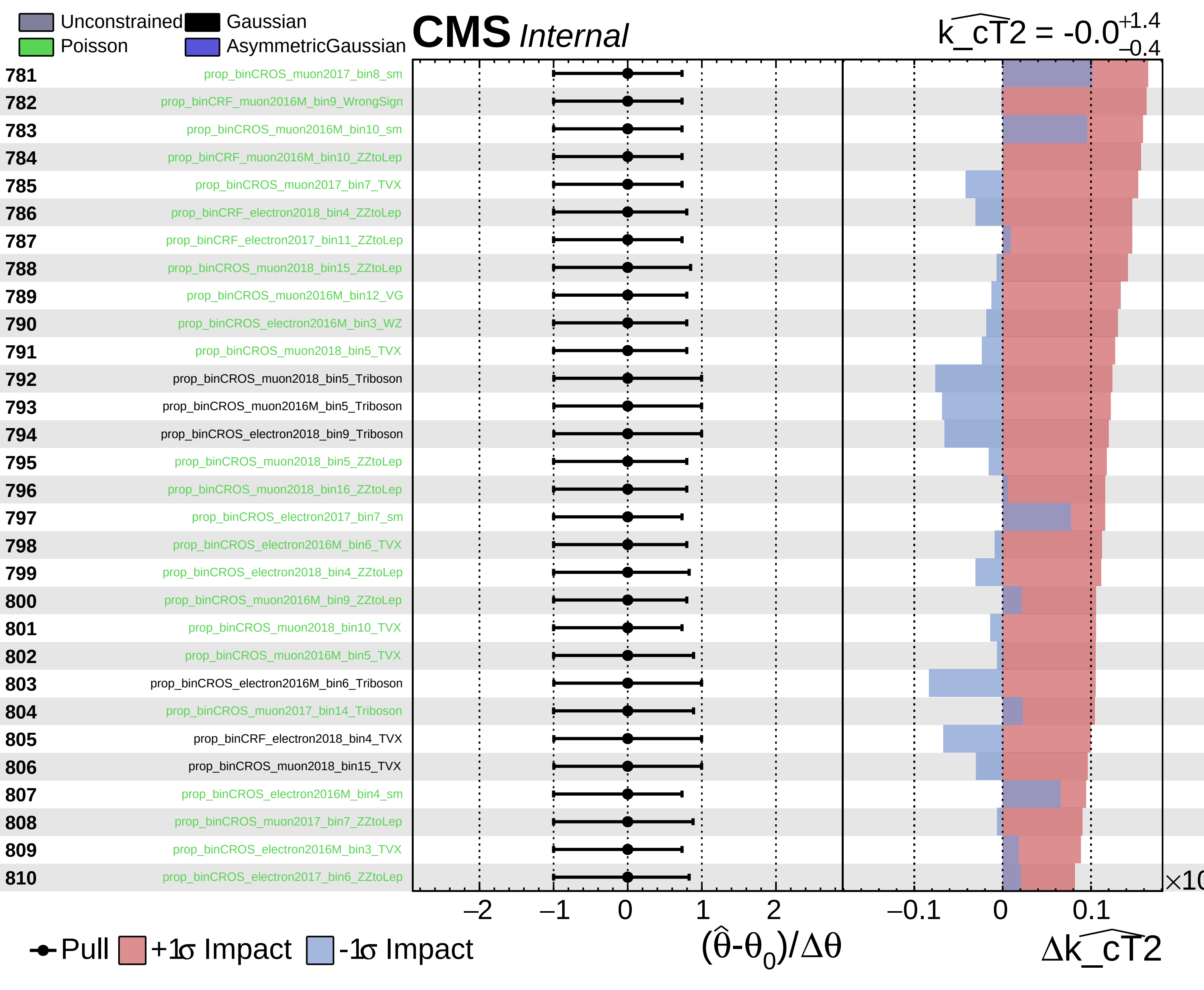
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +1.4 −0.4
691 prop_binSR_electron2016M_bin13_ZZtoLep
692 prop_binCRF_muon2017_bin6_WpWpJJ_QCD
693 prop_binCRF_electron2018_bin13_TVX
694 prop_binCRF_muon2016M_bin8_WrongSign
695 prop_binCRF_electron2017_bin13_Triboson
696 prop_binCRF_electron2016M_bin14_TVX
697 prop_binSR_muon2017_bin10_Triboson
698 prop_binCRF_muon2016M_bin8_TVX
699 prop_binCRF_muon2016M_bin10_WpWpJJ_QCD
700 prop_binCRF_electron2017_bin13_WpWpJJ_QCD
701 prop_binCRF_muon2017_bin7_WpWpJJ_QCD
702 prop_binCRF_electron2016M_bin14_WpWpJJ_QCD
703 prop_binCRF_muon2016M_bin12_WpWpJJ_QCD
704 prop_binCRF_electron2016M_bin13_TVX
705 prop_binCRF_muon2016M_bin13_Triboson
706 prop_binCROS_muon2018_bin5_VG
707 prop_binSR_electron2017_bin10_ZZtoLep
708 prop_binCRF_muon2017_bin12_ZZtoLep
709 prop_binCROS_muon2018_bin17_ZZtoLep
710 prop_binCRF_electron2016M_bin11_Triboson
711 prop_binCRF_muon2017_bin12_Triboson
712 prop_binCROS_electron2017_bin6_TVX
713 prop_binSR_muon2017_bin15_WrongSign
714 prop_binCROS_muon2018_bin17_TVX
715 prop_binCRF_electron2018_bin13_WpWpJJ_QCD
716 prop_binCRF_muon2016M_bin9_WpWpJJ_QCD
717 prop_binCROS_muon2017_bin5_VG
718 prop_binCRF_electron2016M_bin8_ZZtoLep
719 prop_binCRF_electron2017_bin9_ZZtoLep
720 prop_binCRF_electron2017_bin11_WrongSign
−2 −1 0 1 2
−2 0 2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



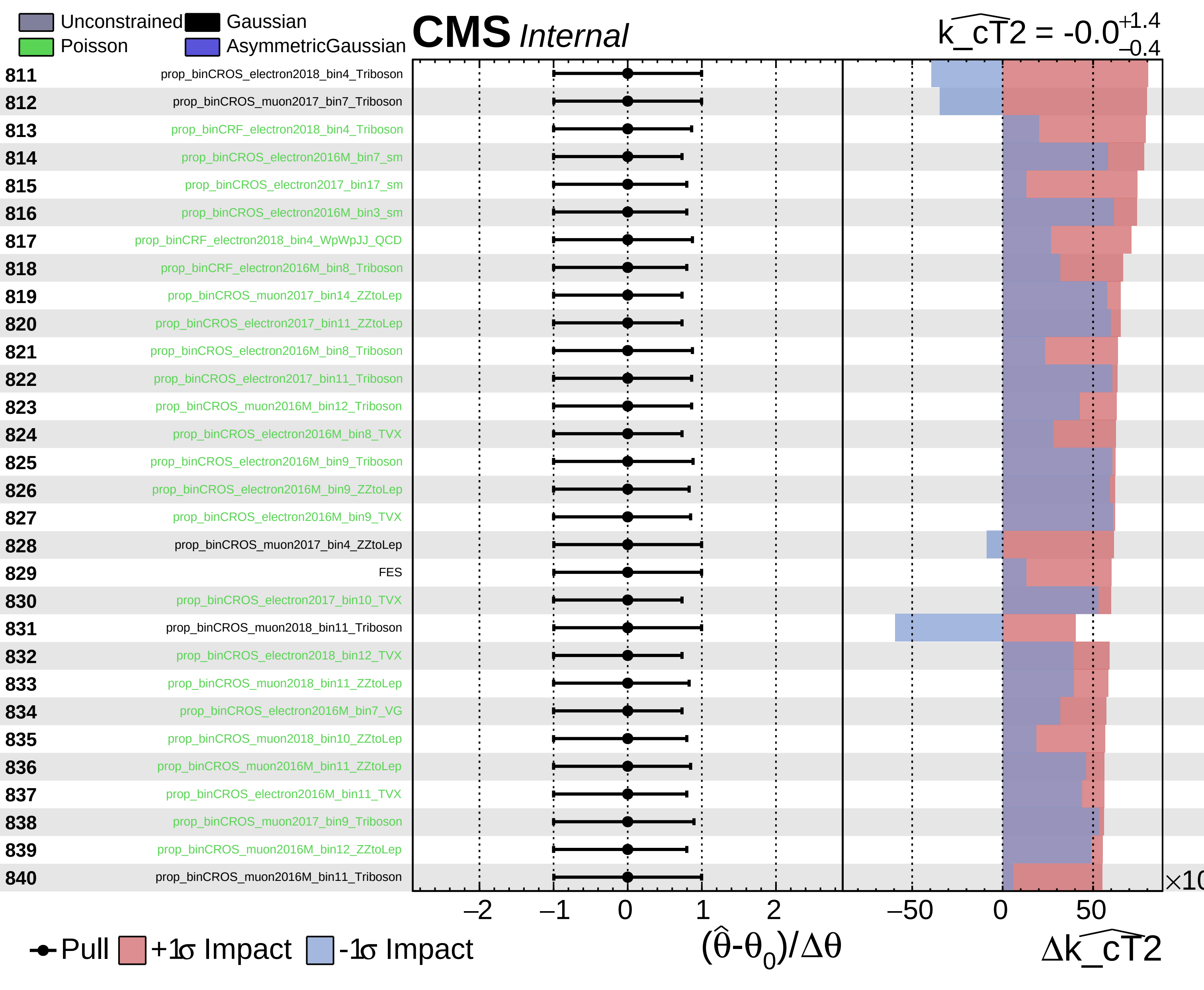
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+1.4}_{-0.4}$
721 prop_binCROS_muon2016M_bin5_WZ
722 prop_binCROS_muon2017_bin17_Triboson
723 prop_binCRF_muon2016M_bin8_WpWpJJ_QCD
724 prop_binCRF_electron2018_bin13_Triboson
725 prop_binCRF_electron2016M_bin12_ZZtoLep
726 prop_binCROS_muon2017_bin10_Triboson
727 prop_binCROS_electron2017_bin17_WrongSign
728 prop_binCRF_electron2016M_bin14_Triboson
729 prop_binSR_muon2016M_bin10_ZZtoLep
730 prop_binCRF_electron2017_bin13_ZZtoLep
731 prop_binSR_electron2017_bin11_WrongSign
732 prop_binCRF_electron2017_bin11_TVX
733 prop_binCROS_muon2016M_bin9_Triboson
734 prop_binCROS_muon2016M_bin5_VG
735 prop_binCROS_muon2016M_bin17_TVX
736 prop_binCROS_muon2018_bin15_Triboson
737 prop_binCRF_electron2016M_bin11_ZZtoLep
738 prop_binCRF_electron2017_bin9_Triboson
739 prop_binCROS_electron2018_bin4_VG
740 prop_binCRF_muon2016M_bin8_Triboson
741 prop_binCRF_electron2016M_bin14_ZZtoLep
742 prop_binCROS_muon2017_bin5_TVX
743 prop_binCRF_electron2016M_bin8_TVX
744 prop_binCROS_muon2016M_bin5_ZZtoLep
745 prop_binCRF_muon2016M_bin10_Triboson
746 prop_binCRF_muon2016M_bin12_Triboson
747 prop_binCRF_electron2017_bin11_Triboson
748 prop_binCROS_muon2018_bin11_sm
749 prop_binCRF_muon2017_bin7_ZZtoLep
750 prop_binCROS_electron2018_bin4_TVX
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



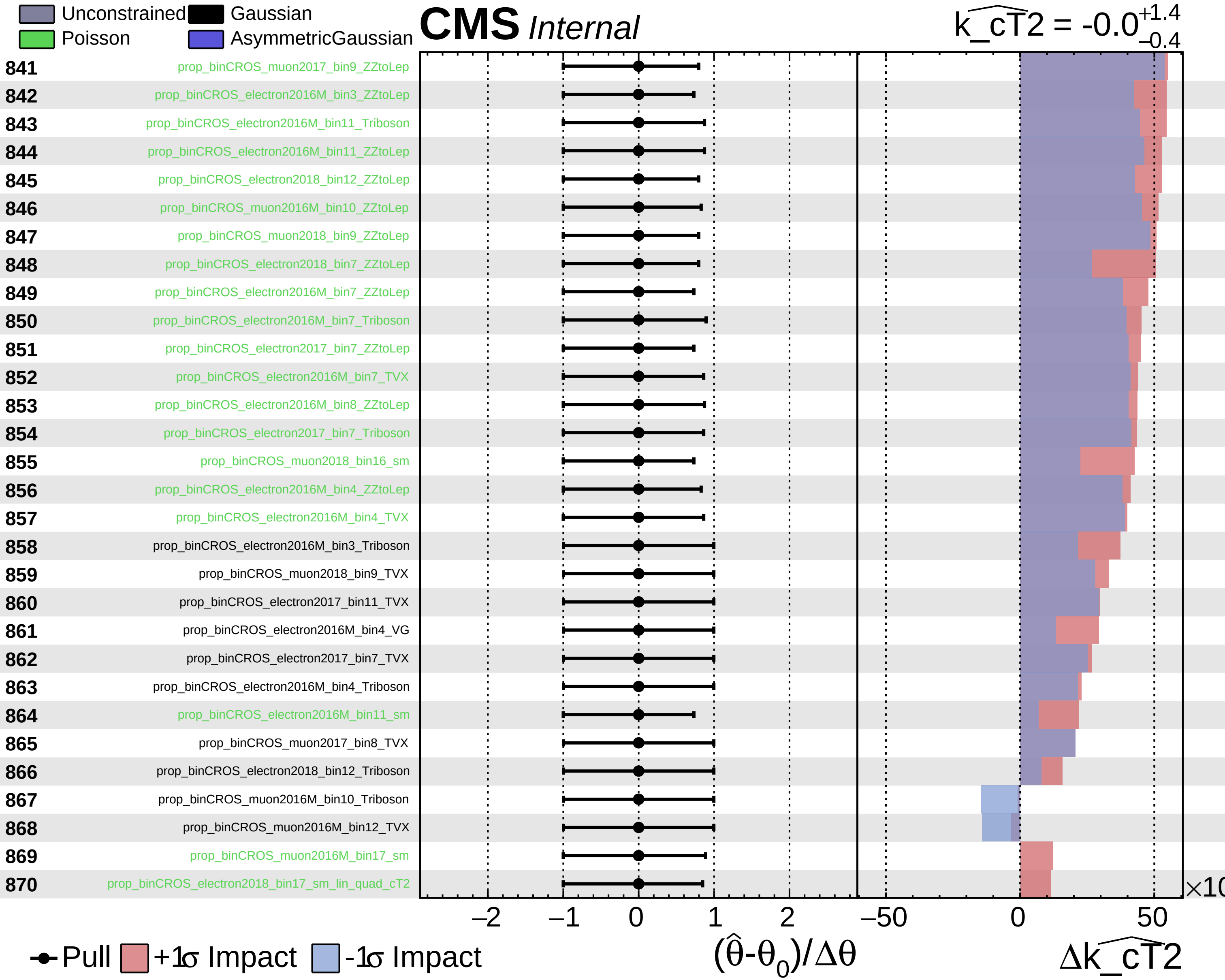
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+1.4}_{-0.4}$
751 prop_binCRF_electron2016M_bin13_ZZtoLep
752 prop_binCROS_electron2018_bin7_TVX
753 prop_binCROS_electron2016M_bin9_sm
754 prop_binCROS_electron2017_bin11_sm
755 prop_binCRF_muon2017_bin7_Triboson
756 prop_binCROS_muon2017_bin5_ZZtoLep
757 prop_binCROS_muon2017_bin17_TVX
758 prop_binCRF_muon2017_bin6_Triboson
759 prop_binSR_muon2017_bin10_ZZtoLep
760 prop_binCROS_electron2018_bin9_TVX
761 prop_binCROS_muon2016M_bin9_sm
762 prop_binCROS_muon2017_bin14_TVX
763 prop_binCROS_muon2017_bin9_sm
764 prop_binCROS_muon2018_bin16_Triboson
765 prop_binCROS_muon2018_bin9_sm
766 prop_binCROS_muon2017_bin10_ZZtoLep
767 prop_binCROS_electron2017_bin6_Triboson
768 prop_binCROS_electron2016M_bin17_ZZtoLep
769 prop_binCROS_muon2017_bin4_Triboson
770 prop_binCROS_electron2016M_bin6_ZZtoLep
771 prop_binCROS_electron2017_bin10_VG
772 prop_binCROS_muon2017_bin5_Triboson
773 prop_binCROS_muon2016M_bin9_TVX
774 prop_binCROS_muon2018_bin10_Triboson
775 prop_binCROS_muon2016M_bin11_TVX
776 prop_binCROS_electron2016M_bin17_sm
777 prop_binCRF_muon2016M_bin8_ZZtoLep
778 prop_binCROS_electron2018_bin9_ZZtoLep
779 prop_binCROS_electron2017_bin10_Triboson
780 prop_binCROS_electron2016M_bin8_sm
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

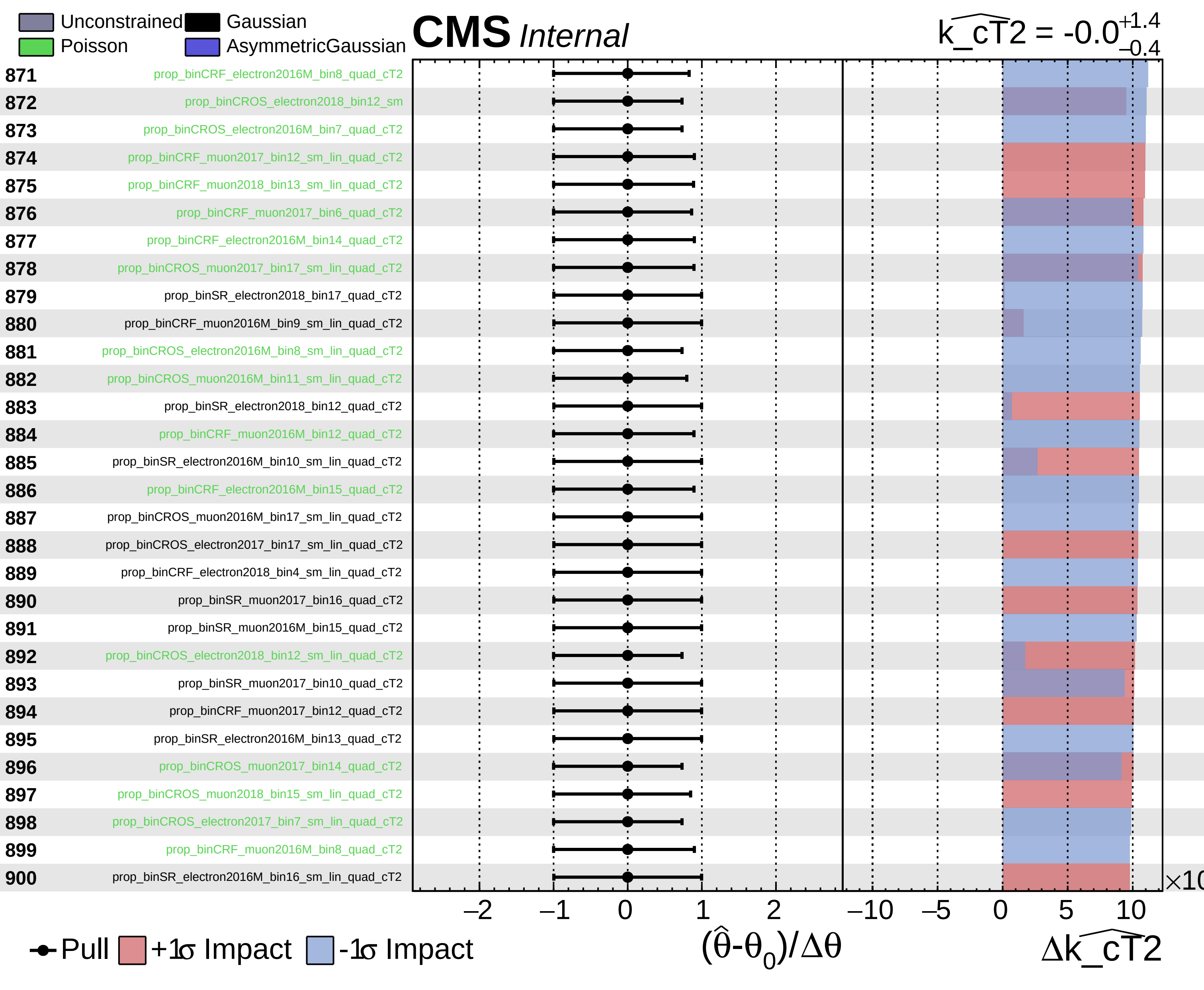


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +1.4 −0.4
781 prop_binCROS_muon2017_bin8_sm
782 prop_binCRF_muon2016M_bin9_WrongSign
783 prop_binCROS_muon2016M_bin10_sm
784 prop_binCRF_muon2016M_bin10_ZZtoLep
785 prop_binCROS_muon2017_bin7_TVX
786 prop_binCRF_electron2018_bin4_ZZtoLep
787 prop_binCRF_electron2017_bin11_ZZtoLep
788 prop_binCROS_muon2018_bin15_ZZtoLep
789 prop_binCROS_muon2016M_bin12_VG
790 prop_binCROS_electron2016M_bin3_WZ
791 prop_binCROS_muon2018_bin5_TVX
792 prop_binCROS_muon2018_bin5_Triboson
793 prop_binCROS_muon2016M_bin5_Triboson
794 prop_binCROS_electron2018_bin9_Triboson
795 prop_binCROS_muon2018_bin5_ZZtoLep
796 prop_binCROS_muon2018_bin16_ZZtoLep
797 prop_binCROS_electron2017_bin7_sm
798 prop_binCROS_electron2016M_bin6_TVX
799 prop_binCROS_electron2018_bin4_ZZtoLep
800 prop_binCROS_muon2016M_bin9_ZZtoLep
801 prop_binCROS_muon2018_bin10_TVX
802 prop_binCROS_muon2016M_bin5_TVX
803 prop_binCROS_electron2016M_bin6_Triboson
804 prop_binCROS_muon2017_bin14_Triboson
805 prop_binCRF_electron2018_bin4_TVX
806 prop_binCROS_muon2018_bin15_TVX
807 prop_binCROS_electron2016M_bin4_sm
808 prop_binCROS_muon2017_bin7_ZZtoLep
809 prop_binCROS_electron2016M_bin3_TVX
810 prop_binCROS_electron2017_bin6_ZZtoLep
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

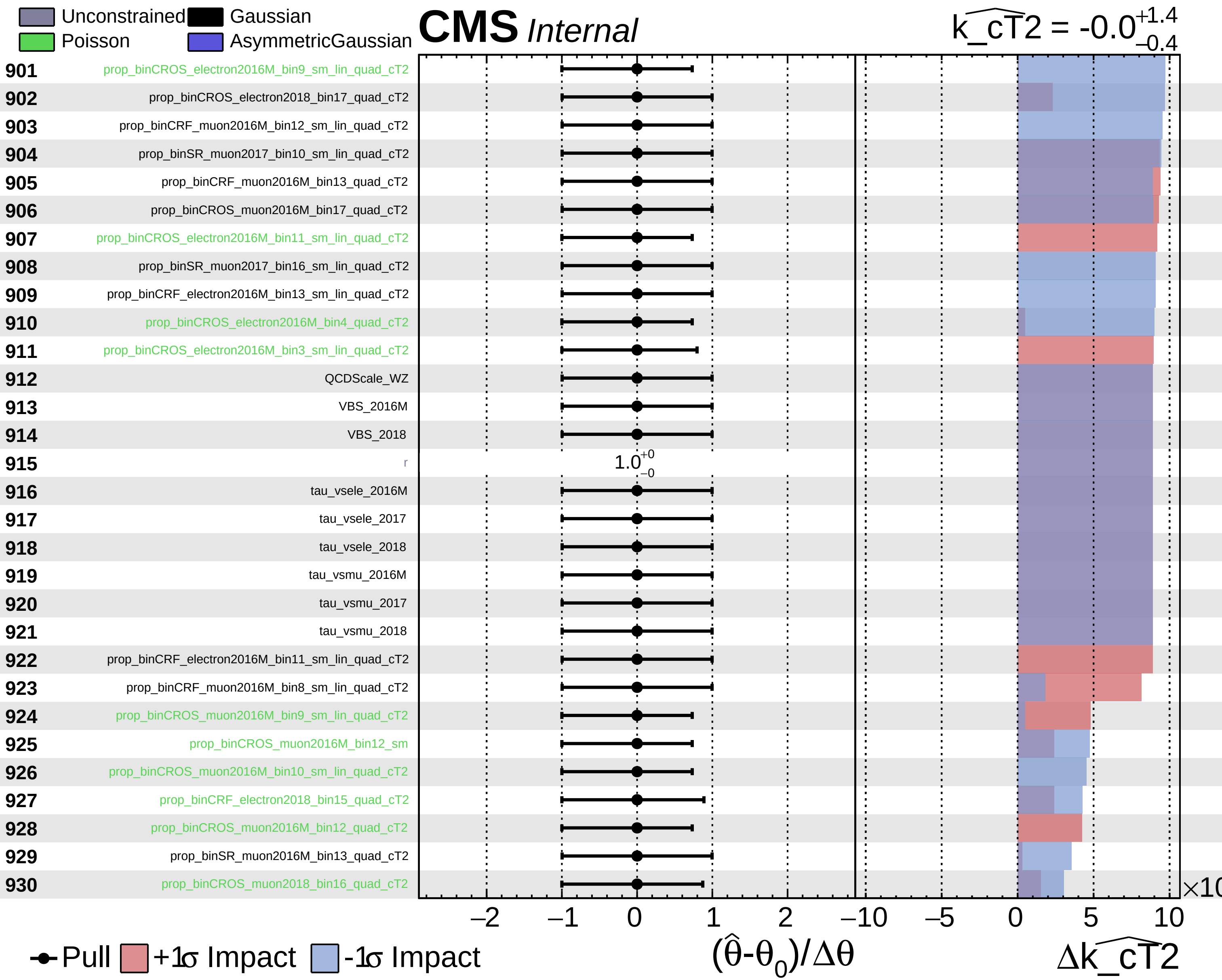


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+1.4}_{-0.4}$
811 prop_binCROS_electron2018_bin4_Triboson
812 prop_binCROS_muon2017_bin7_Triboson
813 prop_binCRF_electron2018_bin4_Triboson
814 prop_binCROS_electron2016M_bin7_sm
815 prop_binCROS_electron2017_bin17_sm
816 prop_binCROS_electron2016M_bin3_sm
817 prop_binCRF_electron2018_bin4_WpWpJJ_QCD
818 prop_binCRF_electron2016M_bin8_Triboson
819 prop_binCROS_muon2017_bin14_ZZtoLep
820 prop_binCROS_electron2017_bin11_ZZtoLep
821 prop_binCROS_electron2016M_bin8_Triboson
822 prop_binCROS_electron2017_bin11_Triboson
823 prop_binCROS_muon2016M_bin12_Triboson
824 prop_binCROS_electron2016M_bin8_TVX
825 prop_binCROS_electron2016M_bin9_Triboson
826 prop_binCROS_electron2016M_bin9_ZZtoLep
827 prop_binCROS_electron2016M_bin9_TVX
828 prop_binCROS_muon2017_bin4_ZZtoLep
829 FES
830 prop_binCROS_electron2017_bin10_TVX
831 prop_binCROS_muon2018_bin11_Triboson
832 prop_binCROS_electron2018_bin12_TVX
833 prop_binCROS_muon2018_bin11_ZZtoLep
834 prop_binCROS_electron2016M_bin7_VG
835 prop_binCROS_muon2018_bin10_ZZtoLep
836 prop_binCROS_muon2016M_bin11_ZZtoLep
837 prop_binCROS_electron2016M_bin11_TVX
838 prop_binCROS_muon2017_bin9_Triboson
839 prop_binCROS_muon2016M_bin12_ZZtoLep
840 prop_binCROS_muon2016M_bin11_Triboson
−2 −1 0 1 2
−50 0 50
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

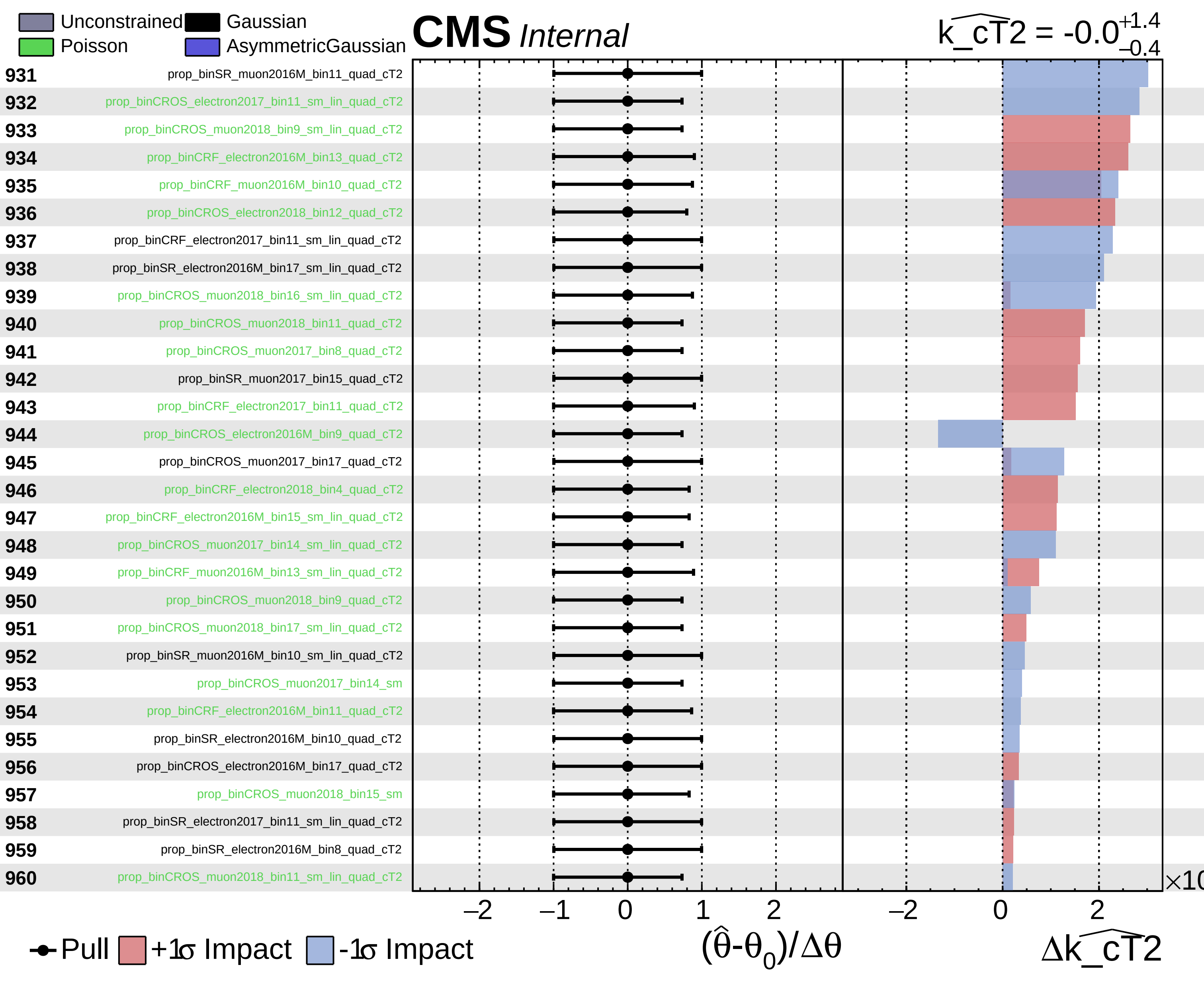

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +1.4 −0.4
841 prop_binCROS_muon2017_bin9_ZZtoLep
842 prop_binCROS_electron2016M_bin3_ZZtoLep
843 prop_binCROS_electron2016M_bin11_Triboson
844 prop_binCROS_electron2016M_bin11_ZZtoLep
845 prop_binCROS_electron2018_bin12_ZZtoLep
846 prop_binCROS_muon2016M_bin10_ZZtoLep
847 prop_binCROS_muon2018_bin9_ZZtoLep
848 prop_binCROS_electron2018_bin7_ZZtoLep
849 prop_binCROS_electron2016M_bin7_ZZtoLep
850 prop_binCROS_electron2016M_bin7_Triboson
851 prop_binCROS_electron2017_bin7_ZZtoLep
852 prop_binCROS_electron2016M_bin7_TVX
853 prop_binCROS_electron2016M_bin8_ZZtoLep
854 prop_binCROS_electron2017_bin7_Triboson
855 prop_binCROS_muon2018_bin16_sm
856 prop_binCROS_electron2016M_bin4_ZZtoLep
857 prop_binCROS_electron2016M_bin4_TVX
858 prop_binCROS_electron2016M_bin3_Triboson
859 prop_binCROS_muon2018_bin9_TVX
860 prop_binCROS_electron2017_bin11_TVX
861 prop_binCROS_electron2016M_bin4_VG
862 prop_binCROS_electron2017_bin7_TVX
863 prop_binCROS_electron2016M_bin4_Triboson
864 prop_binCROS_electron2016M_bin11_sm
865 prop_binCROS_muon2017_bin8_TVX
866 prop_binCROS_electron2018_bin12_Triboson
867 prop_binCROS_muon2016M_bin10_Triboson
868 prop_binCROS_muon2016M_bin12_TVX
869 prop_binCROS_muon2016M_bin17_sm
870 prop_binCROS_electron2018_bin17_sm_lin_quad_cT2
−2 −1 0 1 2
−50 0 50
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



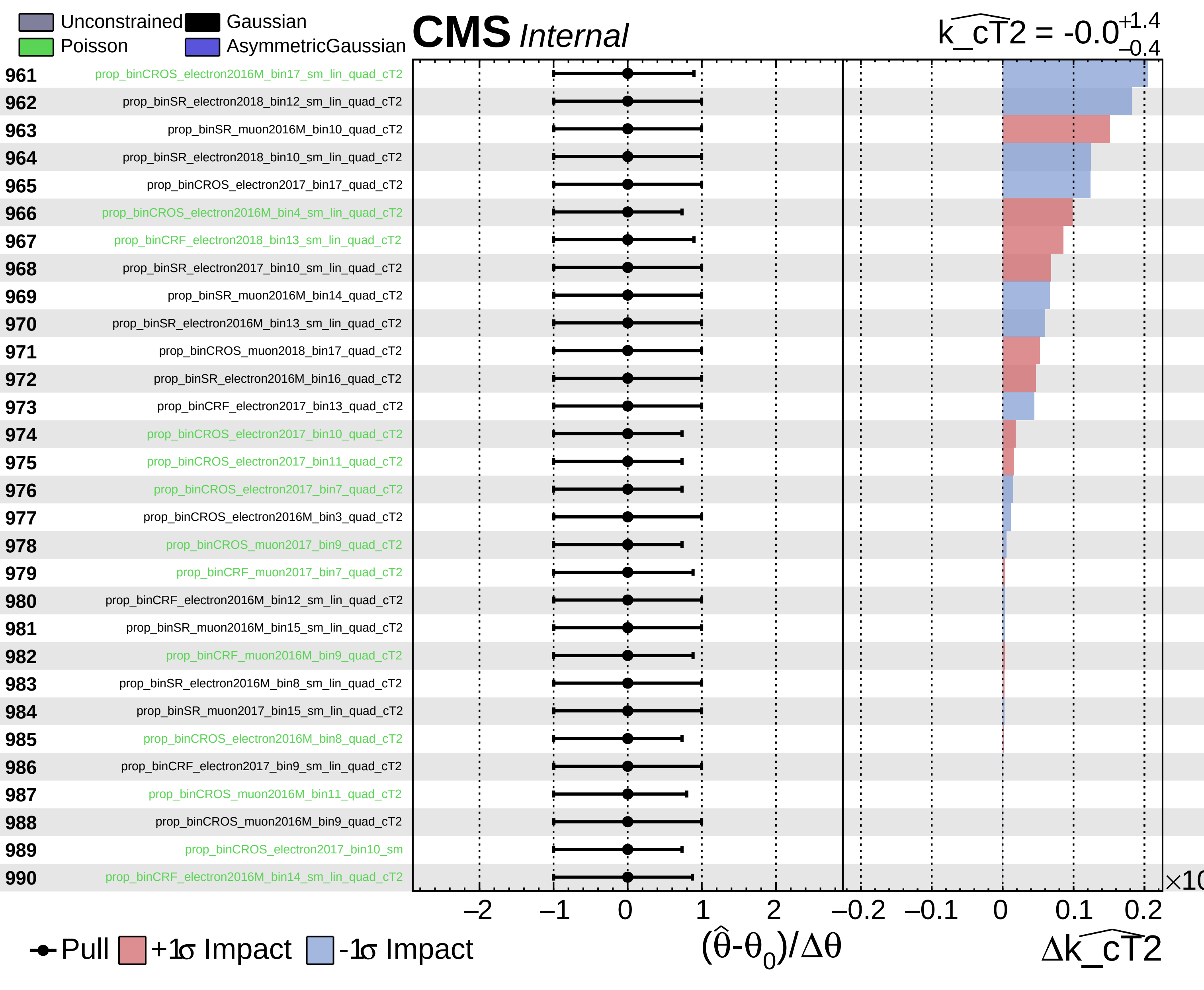
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = -0.0^{+1.4}_{-0.4}$
871 prop_binCRF_electron2016M_bin8_quad_cT2
872 prop_binCROS_electron2018_bin12_sm
873 prop_binCROS_electron2016M_bin7_quad_cT2
874 prop_binCRF_muon2017_bin12_sm_lin_quad_cT2
875 prop_binCRF_muon2018_bin13_sm_lin_quad_cT2
876 prop_binCRF_muon2017_bin6_quad_cT2
877 prop_binCRF_electron2016M_bin14_quad_cT2
878 prop_binCROS_muon2017_bin17_sm_lin_quad_cT2
879 prop_binSR_electron2018_bin17_quad_cT2
880 prop_binCRF_muon2016M_bin9_sm_lin_quad_cT2
881 prop_binCROS_electron2016M_bin8_sm_lin_quad_cT2
882 prop_binCROS_muon2016M_bin11_sm_lin_quad_cT2
883 prop_binSR_electron2018_bin12_quad_cT2
884 prop_binCRF_muon2016M_bin12_quad_cT2
885 prop_binSR_electron2016M_bin10_sm_lin_quad_cT2
886 prop_binCRF_electron2016M_bin15_quad_cT2
887 prop_binCROS_muon2016M_bin17_sm_lin_quad_cT2
888 prop_binCROS_electron2017_bin17_sm_lin_quad_cT2
889 prop_binCRF_electron2018_bin4_sm_lin_quad_cT2
890 prop_binSR_muon2017_bin16_quad_cT2
891 prop_binSR_muon2016M_bin15_quad_cT2
892 prop_binCROS_electron2018_bin12_sm_lin_quad_cT2
893 prop_binSR_muon2017_bin10_quad_cT2
894 prop_binCRF_muon2017_bin12_quad_cT2
895 prop_binSR_electron2016M_bin13_quad_cT2
896 prop_binCROS_muon2017_bin14_quad_cT2
897 prop_binCROS_muon2018_bin15_sm_lin_quad_cT2
898 prop_binCROS_electron2017_bin7_sm_lin_quad_cT2
899 prop_binCRF_muon2016M_bin8_quad_cT2
900 prop_binSR_electron2016M_bin16_sm_lin_quad_cT2
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$



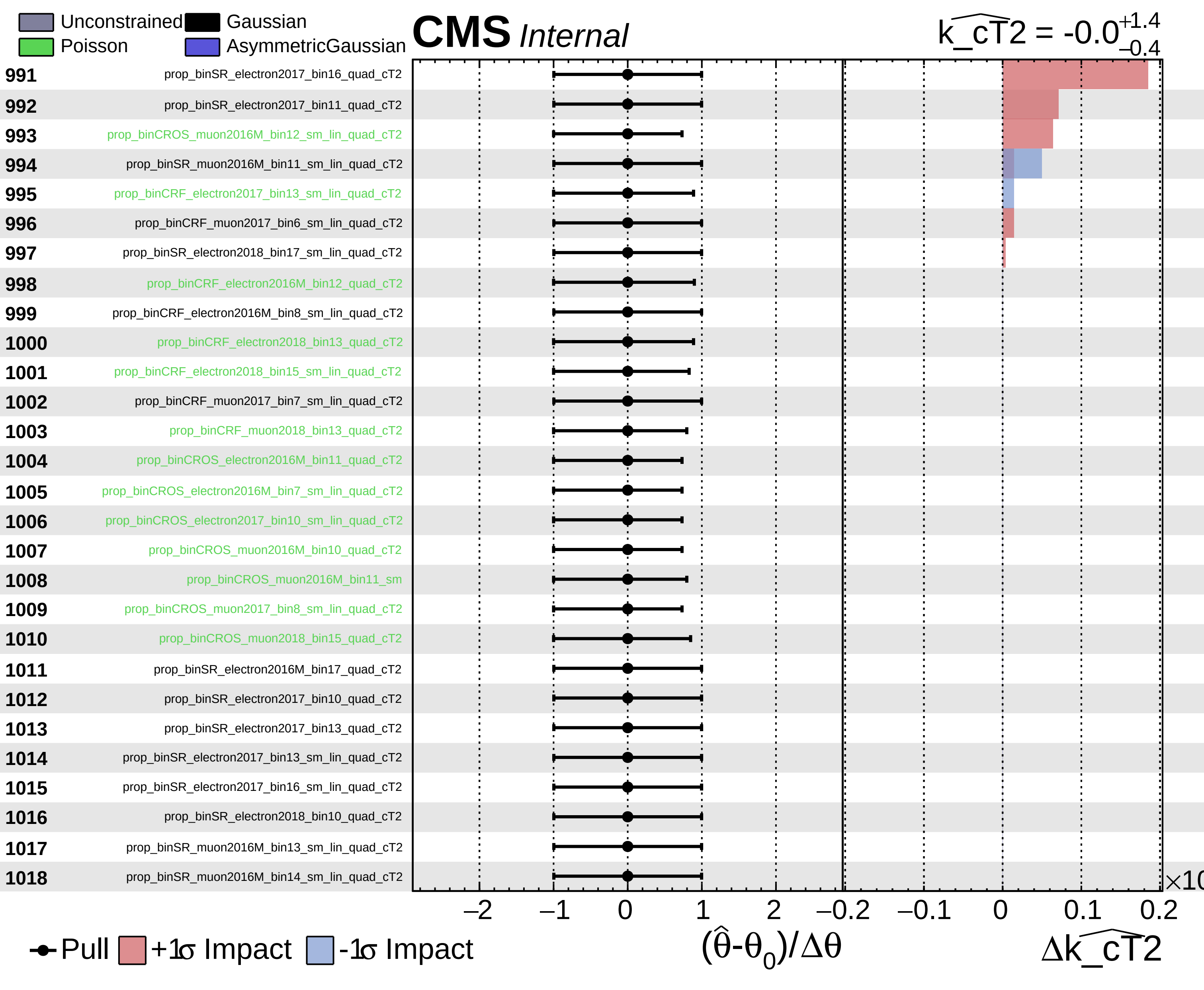
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +1.4 −0.4
901 prop_binCROS_electron2016M_bin9_sm_lin_quad_cT2
902 prop_binCROS_electron2018_bin17_quad_cT2
903 prop_binCRF_muon2016M_bin12_sm_lin_quad_cT2
904 prop_binSR_muon2017_bin10_sm_lin_quad_cT2
905 prop_binCRF_muon2016M_bin13_quad_cT2
906 prop_binCROS_muon2016M_bin17_quad_cT2
907 prop_binCROS_electron2016M_bin11_sm_lin_quad_cT2
908 prop_binSR_muon2017_bin16_sm_lin_quad_cT2
909 prop_binCRF_electron2016M_bin13_sm_lin_quad_cT2
910 prop_binCROS_electron2016M_bin4_quad_cT2
911 prop_binCROS_electron2016M_bin3_sm_lin_quad_cT2
912 QCDScale_WZ
913 VBS_2016M
914 VBS_2018
915 r
1.0 +0 −0
916 tau_vsele_2016M
917 tau_vsele_2017
918 tau_vsele_2018
919 tau_vsmu_2016M
920 tau_vsmu_2017
921 tau_vsmu_2018
922 prop_binCRF_electron2016M_bin11_sm_lin_quad_cT2
923 prop_binCRF_muon2016M_bin8_sm_lin_quad_cT2
924 prop_binCROS_muon2016M_bin9_sm_lin_quad_cT2
925 prop_binCROS_muon2016M_bin12_sm
926 prop_binCROS_muon2016M_bin10_sm_lin_quad_cT2
927 prop_binCRF_electron2018_bin15_quad_cT2
928 prop_binCROS_muon2016M_bin12_quad_cT2
929 prop_binSR_muon2016M_bin13_quad_cT2
930 prop_binCROS_muon2018_bin16_quad_cT2
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +1.4 −0.4
931 prop_binSR_muon2016M_bin11_quad_cT2
932 prop_binCROS_electron2017_bin11_sm_lin_quad_cT2
933 prop_binCROS_muon2018_bin9_sm_lin_quad_cT2
934 prop_binCRF_electron2016M_bin13_quad_cT2
935 prop_binCRF_muon2016M_bin10_quad_cT2
936 prop_binCROS_electron2018_bin12_quad_cT2
937 prop_binCRF_electron2017_bin11_sm_lin_quad_cT2
938 prop_binSR_electron2016M_bin17_sm_lin_quad_cT2
939 prop_binCROS_muon2018_bin16_sm_lin_quad_cT2
940 prop_binCROS_muon2018_bin11_quad_cT2
941 prop_binCROS_muon2017_bin8_quad_cT2
942 prop_binSR_muon2017_bin15_quad_cT2
943 prop_binCRF_electron2017_bin11_quad_cT2
944 prop_binCROS_electron2016M_bin9_quad_cT2
945 prop_binCROS_muon2017_bin17_quad_cT2
946 prop_binCRF_electron2018_bin4_quad_cT2
947 prop_binCRF_electron2016M_bin15_sm_lin_quad_cT2
948 prop_binCROS_muon2017_bin14_sm_lin_quad_cT2
949 prop_binCRF_muon2016M_bin13_sm_lin_quad_cT2
950 prop_binCROS_muon2018_bin9_quad_cT2
951 prop_binCROS_muon2018_bin17_sm_lin_quad_cT2
952 prop_binSR_muon2016M_bin10_sm_lin_quad_cT2
953 prop_binCROS_muon2017_bin14_sm
954 prop_binCRF_electron2016M_bin11_quad_cT2
955 prop_binSR_electron2016M_bin10_quad_cT2
956 prop_binCROS_electron2016M_bin17_quad_cT2
957 prop_binCROS_muon2018_bin15_sm
958 prop_binSR_electron2017_bin11_sm_lin_quad_cT2
959 prop_binSR_electron2016M_bin8_quad_cT2
960 prop_binCROS_muon2018_bin11_sm_lin_quad_cT2
−2
−1
0
1
2
−2
0
2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT2



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0+1.4−0.4
961 prop_binCROS_electron2016M_bin17_sm_lin_quad_cT2
962 prop_binSR_electron2018_bin12_sm_lin_quad_cT2
963 prop_binSR_muon2016M_bin10_quad_cT2
964 prop_binSR_electron2018_bin10_sm_lin_quad_cT2
965 prop_binCROS_electron2017_bin17_quad_cT2
966 prop_binCROS_electron2016M_bin4_sm_lin_quad_cT2
967 prop_binCRF_electron2018_bin13_sm_lin_quad_cT2
968 prop_binSR_electron2017_bin10_sm_lin_quad_cT2
969 prop_binSR_muon2016M_bin14_quad_cT2
970 prop_binSR_electron2016M_bin13_sm_lin_quad_cT2
971 prop_binCROS_muon2018_bin17_quad_cT2
972 prop_binSR_electron2016M_bin16_quad_cT2
973 prop_binCRF_electron2017_bin13_quad_cT2
974 prop_binCROS_electron2017_bin10_quad_cT2
975 prop_binCROS_electron2017_bin11_quad_cT2
976 prop_binCROS_electron2017_bin7_quad_cT2
977 prop_binCROS_electron2016M_bin3_quad_cT2
978 prop_binCROS_muon2017_bin9_quad_cT2
979 prop_binCRF_muon2017_bin7_quad_cT2
980 prop_binCRF_electron2016M_bin12_sm_lin_quad_cT2
981 prop_binSR_muon2016M_bin15_sm_lin_quad_cT2
982 prop_binCRF_muon2016M_bin9_quad_cT2
983 prop_binSR_electron2016M_bin8_sm_lin_quad_cT2
984 prop_binSR_muon2017_bin15_sm_lin_quad_cT2
985 prop_binCROS_electron2016M_bin8_quad_cT2
986 prop_binCRF_electron2017_bin9_sm_lin_quad_cT2
987 prop_binCROS_muon2016M_bin11_quad_cT2
988 prop_binCROS_muon2016M_bin9_quad_cT2
989 prop_binCROS_electron2017_bin10_sm
990 prop_binCRF_electron2016M_bin14_sm_lin_quad_cT2
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = -0.0 +1.4 −0.4
991 prop_binSR_electron2017_bin16_quad_cT2
992 prop_binSR_electron2017_bin11_quad_cT2
993 prop_binCROS_muon2016M_bin12_sm_lin_quad_cT2
994 prop_binSR_muon2016M_bin11_sm_lin_quad_cT2
995 prop_binCRF_electron2017_bin13_sm_lin_quad_cT2
996 prop_binCRF_muon2017_bin6_sm_lin_quad_cT2
997 prop_binSR_electron2018_bin17_sm_lin_quad_cT2
998 prop_binCRF_electron2016M_bin12_quad_cT2
999 prop_binCRF_electron2016M_bin8_sm_lin_quad_cT2
1000 prop_binCRF_electron2018_bin13_quad_cT2
1001 prop_binCRF_electron2018_bin15_sm_lin_quad_cT2
1002 prop_binCRF_muon2017_bin7_sm_lin_quad_cT2
1003 prop_binCRF_muon2018_bin13_quad_cT2
1004 prop_binCROS_electron2016M_bin11_quad_cT2
1005 prop_binCROS_electron2016M_bin7_sm_lin_quad_cT2
1006 prop_binCROS_electron2017_bin10_sm_lin_quad_cT2
1007 prop_binCROS_muon2016M_bin10_quad_cT2
1008 prop_binCROS_muon2016M_bin11_sm
1009 prop_binCROS_muon2017_bin8_sm_lin_quad_cT2
1010 prop_binCROS_muon2018_bin15_quad_cT2
1011 prop_binSR_electron2016M_bin17_quad_cT2
1012 prop_binSR_electron2017_bin10_quad_cT2
1013 prop_binSR_electron2017_bin13_quad_cT2
1014 prop_binSR_electron2017_bin13_sm_lin_quad_cT2
1015 prop_binSR_electron2017_bin16_sm_lin_quad_cT2
1016 prop_binSR_electron2018_bin10_quad_cT2
1017 prop_binSR_muon2016M_bin13_sm_lin_quad_cT2
1018 prop_binSR_muon2016M_bin14_sm_lin_quad_cT2
×10
−2 −1 0 1 2 −0.2 −0.1 0 0.1 0.2
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2